DSGE Model Estimation via VAR IRF Matching: The Short and Long of It *

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Abstract

Following De Graeve and Westermark (2025) and Baumeister (2025), I increase the lag length of the VAR in Altig et al. (2005) to re-estimate their DSGE. Moving from $p = 4$ to $p = 12$ lags, I find that both the VAR and DSGE impulse responses, and the implied structural parameters, change substantially. The Calvo price parameters that distinguish the observationally equivalent firm-specific and homogeneous capital versions of the model converge to one another, from 0.34 against 0.82 at $p = 4$ to 0.89 against 0.90 at $p = 12$. The implied duration of price contracts in the firm-specific model rises from 1.5 to 9.0 quarters, equal to that of the homogeneous capital model, re-opening the micro–macro discrepancy in price stickiness that firm-specific capital was introduced to resolve.

Keywords: Truncation Bias, Long-Lag VARs, DSGE Estimation, IRF Minimum Distance , Nominal Rigidities
JEL Codes: C32, C52, E32

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1 Introduction

1.1 Motivation

Inflation is persistent. Once it rises it tends to stay high for some time, and bringing it back down is slow. This is a stylised fact of the data, and it is visible in the post second World War history of US inflation reported in Figure 1.

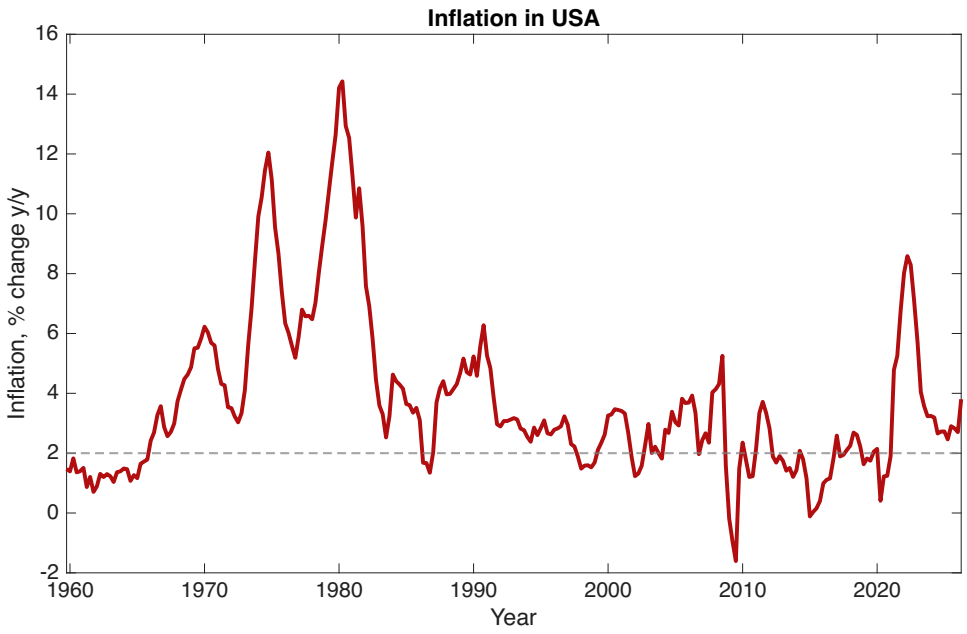


Figure 1: Inflation in the US, 1959Q2–2026Q2: year on year growth rate of the Consumer Price Index, in percent.

Explaining it is one of the central tasks of monetary business cycle research, and it requires knowing how the economy responds to shocks. AGGIUSTARE STA FRASE Since the systematic component of policy is endogenous, this means isolating the non systematic part, the monetary policy shock, small and infrequent as it is (Ravenna, 2007; Aruoba and Drechsel, 2024).

Dynamic Stochastic General Equilibrium (DSGE) models account for inflation inertia only by requiring firms to re-optimise their prices rarely, every six quarters and, in versions without indexation to past inflation, even every two years. The microeconometric evidence finds price changes considerably more frequent (Bils and Klenow, 2004; Golosov and Lucas, 2007; Klenow and Kryvtsov, 2008), so aggregate and micro evidence point in opposite directions. Altig et al. (2011) propose a resolution. They build two versions of the Christiano et al. (2005) model, one with homogeneous and one with firm-specific capital, and in the second inflation is inertial even though firms re-optimise on average every 1.5 quarters, because the firms that do reset their price move it by little rather than rarely.

Both versions are taken to the data by matching impulse responses. Impulse Response Function Matching Estimation (IRFME) is a limited information strategy that chooses the structural parameters minimising the distance between the responses the model implies and those estimated from an identified VAR (Christiano et al., 2005; Altig et al., 2011; Rossi et al., 2012). The strategy is attractive precisely because it is partial: the researcher takes a stand on the identification of a subset of shocks and stays agnostic about the rest of the data.

That attractiveness rests on one maintained assumption, that the VAR delivers consistent estimates of the moments being matched. If it is misspecified, the specification error propagates into the structural statistics, at every horizon that enters the objective function (Ramey, 2016; Canova and Fosso, 2025). Altig et al. (2011, fn. 10, p. 3) say as much themselves: "the quality of our estimation strategy depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks".

One source of misspecification is the lag order. Truncation bias arises whenever the VAR carries fewer lags than its data generating process, and it distorts the autoregressive coefficients and the covariance matrix of the residuals alike, hence the responses computed from them. Lag length, however, is rarely treated as a substantive modelling decision: it is settled by forecasting criteria or by convention, four lags for quarterly data, and a recent literature argues that neither is the right instrument for a structural question (De Graeve and Westermark, 2025; Baumeister, 2025). This thesis asks what that choice does to the structural conclusions drawn from such a VAR in the contest of DSGE estimation.

1.2 Contribution and Results

I replicate Altig et al. (2005)[1] and re-estimate their DSGE by IRF matching against VARs of increasing lag order, from the baseline $p = 4$ to $p = 8$ and $p = 12$, one year of dynamics against two and three. The question is what lag truncation does to DSGE structural parameter estimates, and not only to the impulse responses from which they are obtained. To the best of my knowledge this is the first study to ask it.

The exercise extends De Graeve and Westermark (2025) along three dimensions. Their evidence comes from bivariate systems generated by DSGE models with an exact VAR(∞) representation, so the truncated VAR is nested in a known truth. The VAR of Altig et al.

[1]The replication code made available from the authors is NBER working paper from 2005. Hence, I replicate and refer to those results.

(2011) confronts three shocks with ten observable variables, so it is stochastically singular and no such representation exists. The system is also expensive to parametrise: at $p = 12$ it carries 1210 coefficients, 121 per equation, and 38 degrees of freedom survive out of 159 usable observations, with the limit at $p = 16$. Tripling the lag length is therefore the largest perturbation the data permit, and a demanding test of their claim. The estimation is also frequentist throughout, so the parametrisation is not disciplined by a prior.

The second contribution is methodological and concerns the choice of p. Baumeister (2025) argues for long lags when the question is itself about the long run, and Antolín-Díaz and Surico (2025) set $p = h$ by design to draw inference at horizon h. My point does not depend on the horizon of the question. When a DSGE is estimated by IRF matching the objective function reaches horizon h by construction, $h = 20$ quarters in Altig et al. (2011), so a misspecification of the dynamics enters the minimisation at every one of those horizons. Even when the substantive question is about the short run, then, p has to be chosen with h in mind.

The VAR carries a short run recursive restriction for the monetary policy shock and long run restrictions for the two technology shocks, which makes it an unusually clean setting in which to separate the two channels of Ravenna (2007); pure truncation bias and identification bias. The responses to a monetary policy shock keep their sign and their shape at every lag order and move only in timing, with peaks arriving about a year later at $p = 12$. The responses to the two technology shocks do not: they change level and, for several variables, sign, and they disagree already on impact, where the relative price of investment falls from about -0.45 to -0.1 on an embodied technology shock. Pure truncation distorts propagation and cannot by itself produce a disagreement at horizon zero, so this is the trace Ravenna (2007) attributes to identification bias, and it appears exactly where the theory says it should.

Confidence intervals do not explode as the lag length grows. At $p = 12$ the bootstrap bands are narrower than at the baseline at every horizon and for every variable, with the single exception of MZM growth, so the variance reduction documented by De Graeve and Westermark (2025) operates at the parametrisation frontier. It reaches the structural estimates too: the Phillips curve slope γ, which cannot be distinguished from zero at the baseline, is six times more precise at $p = 12$.

The economic result follows from a single parameter. The cost of varying capital utilisation σ_a falls from 1.9938 at the baseline to 0.0042 at $p = 12$, and σ_a is what makes firm specificity bind. With utilisation almost free to vary, a firm tied to its own capital can still obtain the services it needs, its marginal cost stops rising with its own output,

and the real rigidity χ rises from 0.03 to 0.87 on non overlapping simulated intervals. The implied duration of price contracts, 1.5 quarters under firm-specific capital against 5.6 under homogeneous capital at the baseline, becomes 9.0 and 9.6 quarters at $p = 12$. Firm specificity survives as an assumption about ownership with no economic content, the two versions of the model say the same thing about price stickiness, and the reconciliation with the micro evidence is gone.

1.3 Related Literature

Long-lag VARs. This work builds directly on De Graeve and Westermark (2025), and on their earlier De Graeve and Westermark (2013), who show that increasing the lag length reduces misspecification and with it the variance of the residuals, so that uncertainty need not grow with the parametrisation. Baumeister (2025) makes the same case for questions about the long run, and Antolín-Díaz and Surico (2025) set $p = h$ by design. Plagborg-Møller and Wolf (2021) establish the asymptotic properties that make long lag specifications defensible, while González-Casasús and Schorfheide (2025) propose a lag selection criterion built for structural inference rather than for forecasting. The same trade off separates VARs from local projections (Li et al., 2024; Montiel-Olea et al., 2025; Ludwig, 2026), voglio citare un po' di più Ludwig che è l'evoluzione di P-M&W2021 trova la mappatura di LP and VARs in samples and breake down the li montiel olea et al 2025 "lp vs VAS .." which claims that LP are best compared to vars in everything. and it connects to the reappraisal of the bias variance trade off in richly parametrised models (Belkin et al., 2019; Antolín-Díaz et al., 2026). posso citare antolin diaz and co su double desent phenomena? anche se c'è solo solo le slides?

Truncation bias and the DSGE–VAR mapping. Fernández-Villaverde et al. (2007) give the conditions under which the solution of a DSGE admits a VAR(∞) representation, and Ravenna (2007) separates the two channels through which truncating it operates. Long run identification is the fragile case (Faust and Leeper, 1997; Erceg et al., 2005; Christiano et al., 2007; Chari et al., 2008), and the same fragility motivates the debate on fundamentalness and on how much of it is really needed (Lippi and Reichlin, 1993; Forni et al., 2016).

Impulse response matching and DSGE estimation. The estimator goes back to Rotemberg and Woodford (1997) and is used by Christiano et al. (2005), Iacoviello (2005), Boivin and Giannoni (2006) and Altig et al. (2011). Rossi et al. (2012) study its asymptotic properties, Giacomini (2013) and Theodoridis (2011) discuss what is gained and lost relative to full information methods, and Canova (1994, 2007) set out the wider case for judging a model by economic rather than statistical distance.

1.4 Outline

Section 2 presents the model, the micro–macro conflict it is meant to resolve, the role of the real rigidity χ and of the cost of varying capital utilisation σ_a, and the IRF matching estimator. Section 3 sets out truncation bias, the two channels through which it reaches structural estimates, and the case for long-lag VARs. Section 4 reports the VAR impulse responses across lag orders and the behaviour of the confidence intervals. Section 5 reports the structural estimates, the real rigidity, the Calvo price stickiness parameters and the implied duration of prices. Section 6 concludes. The appendices collect the data, the identification strategy, the derivation of the model and the additional results.

2 The Theoretical Model

2.1 Inflation inertia and the micro–macro conflict

Altig et al. (2011) start from a tension in the New Keynesian literature. To account for inertia in inflation, standard models require firms to re-optimise prices infrequently; every six quarters, and in versions without indexation to past inflation even every two years. This sits uneasily with the microeconometric evidence, which finds price changes considerably more frequent Bils and Klenow (2004); Golosov and Lucas (2007); Klenow and Kryvtsov (2008).

Altig et al. (2005) propose a resolution. In their model inflation is inertial even though firms re-optimise on average every 1.5 quarters. Inertia arises not because firms change prices rarely, but because when they do it, they change them by a small amount: each firm's short-run marginal cost is increasing in its own output, which dampens the optimal price adjustment. The underlying assumption that drives this result is that capital is predetermined within the period and firm-specific, so that factor of production cannot be instantly and costlessly reallocated across firms.

2.2 Homogeneous and firm-specific capital

Altig et al. (2011) build on the Christiano et al. (2005) model, adding the two technology shocks, and construct two versions of it: one in which capital is homogeneous, as in standard formulations, and one in which it is firm-specific.

Once log-linearised, the only difference between the two sets of equilibrium conditions lies in the equation relating inflation π_t to marginal cost, the New Keynesian Phillips curve,

$$\Delta\hat{\pi}_t = \mathbb{E}\big[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t \mid \Omega_t\big], \tag{1}$$

where $\hat{s}_t$ is economy-wide average real marginal cost in units of the final good, and the slope γ maps into the structural parameters as

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\,\chi. \tag{2}$$

The information set Ω_t contains the current realisation of the technology shocks but not that of the monetary policy innovation. Here ξ_p is the price stickiness parameter of the Calvo (1983) pricing scheme, so that a fraction $1-\xi_p$ of firms reset their price each quarter and the implied duration of a price contract is $D = 1/(1-\xi_p)$.

Both models are parametrised in terms of γ rather than ξ_p. Given the remaining parameters they are then observationally equivalent with respect to aggregate prices and quantities, so that no stand need be taken on which version is being estimated. What distinguishes them is the value of ξ_p that a given γ implies, and that in turn depends on χ.

The real rigidity parameter χ measures the strategic complementarity in price setting of Ball and Romer (1990). It works as an attenuation factor, takes values in $(0, 1]$, and the closer it lies to zero the stronger the attenuation: a firm that is able to reset its price moves it by less, for a given movement in economy-wide marginal cost. In the *homogeneous* capital model χ is set to be one and the attenuation is absent. In the *firm-specific* model χ is a non-linear function of the structural parameters, namely of the calibrated $[\alpha, \beta, \delta, \lambda_f]$, respectively the capital share in the Cobb-Douglas production function, the discount factor, the depreciation rate and the average price markup, and of two estimated parameters, $[\sigma_a, S'']$, capital utilisation cost and the investment adjustment cost respectively.

The investment adjustment cost S'' enters χ too, but through a different channel: it governs the cost of changing the flow of new investment, which shapes how quickly the firm's own capital stock accumulates in future periods, not how much capital it can access this quarter. Since χ is pinned down mainly by the within-period margin that σ_a governs, S'' has correspondingly little quantitative bearing on it.

The role of capital utilisation cost σ_a is best understood by following what happens after an unexpected, positive demand shock. Aggregate demand rises within the quarter and firms have to produce more. Hours expand first, and wages are themselves set on a Calvo scheme, so only the $1-\xi_w$ of households that can reoptimise choose a new wage, while the rest adjust only mechanically through indexation. The aggregate nominal wage therefore rises, and the real wage with it, but by little. Capital is another matter, and it is here that the two models part company.

Under homogeneous capital there is a competitive economy-wide market for capital services. In the aggregate the capital stock is predetermined within the quarter, so the additional services can only come from higher utilisation, and how far the rental rate must rise to call them forth is governed by σ_a. That much is true in both models, and it is not what separates them. But in the homogeneous capital model each takes the rental rate as given and hires as many capital services as it wants at it, since capital can be moved across firms within the period. Any single firm's capital stock is therefore not predetermined, even though the economy's is. Altig et al. (2005, p. 2) put the assumption plainly: "a firm's capital stock is not pre-determined and all factors of production, including capital, can be instantaneously transferred across firms, without any cost, in perfectly competitive markets". Both factors are then available at prices the firm does not influence, so its marginal cost is the same as every other firm's and does not depend on how much it produces itself. That is what $\chi = 1$ records.

Under the firm-specific capital model there is no market for capital services. The firm is left with the capital it already owns, which cannot be reallocated within the period, so the only way to obtain more capital services is to run its own stock harder. It would be willing to pay for additional services, but there is no market on which to buy them, and σ_a governs how much raising utilisation costs it instead. When σ_a is high that margin is expensive, so most of the additional output has to come from adding hours to a quantity of capital services that barely moves. The marginal product of labour falls as the fixed factor binds, and the firm's own marginal cost rises with its own output.

This changes the decision of the optimiser firms that can reset their price. Economy-wide marginal cost has risen, so those firms wish to raise prices. But a firm that raises its price by more than the others sees its relative price rise, its demand fall, and its own output fall with it, and under firm-specific capital a lower own output means a lower own marginal cost, which argues for a smaller increase in the first place. Having internalised that producing more is what raises its costs, the firm moves its price by less than it would in the absence of this real rigidity, the more so the steeper its own marginal cost curve and the more elastic the demand it faces, since a given price increase then costs it more output. Inflation responds less to a given movement in economy-wide marginal cost, and this is precisely what a χ below one means.

It is worth being precise about what is doing the work here, because σ_a is a single parameter, estimated once, which does not take different values across the two models. What differs is whether it reaches χ at all. Under homogeneous capital the market for capital services severs the link between a firm's own output and its own marginal cost, and $\chi = 1$ whatever σ_a happens to be. Under firm-specific capital the firm is tied to the

capital it already owns, and σ_a decides how binding that tie is.

The limiting case makes the point. If σ_a is close to zero, utilisation is almost free to vary, so a firm tied to its own capital can still draw from it whatever volume of capital services it needs, at negligible cost. Owning the capital rather than renting it then makes no practical difference: the firm's own marginal cost is as flat as it would be with a market for capital services, χ converges to one, and the firm-specific model collapses onto the homogeneous one. Firm specificity is an assumption about ownership, but its economic content comes entirely from the cost of varying utilisation, and without that cost the constraint exists in name only.

Since χ is decreasing in σ_a and γ is proportional to χ, a larger σ_a lowers γ for any given ξ_p. Reading the Phillips curve slope γ, equation (2), in the other direction is what matters for estimation, as γ is estimated from the data. The term $(1-\xi_p)(1-\beta\xi_p)/\xi_p$ is decreasing in ξ_p, so holding γ fixed, a smaller χ must be offset by a larger value of that term, which requires a smaller ξ_p. For a given slope of the Phillips curve, then, ξ_p is a decreasing function of σ_a.

This is what resolves the micro macro price conflict. The empirical literature finds the Phillips curve to be flat, that is γ small. In the homogeneous capital model $\chi = 1$, so a small γ can only be delivered by pushing ξ_p close to one, and the implied duration $D = 1/(1-\xi_p)$ becomes long enough to sit in open conflict with the micro literature evidence. In the firm-specific model the same small γ is delivered instead by a χ well below one, which leaves room for a much lower ξ_p. Inflation is inertial not because firms reset their prices rarely, but because when they do they move them by little. This is how Altig et al. (2005) obtain an average duration of 1.5 quarters and still match the aggregate evidence on inflation.

The resolution, however, rests on a single estimated parameter, capital utilisation cost σ_a, and the authors acknowledge the fragility themselves: "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p" (Altig et al., 2011, p. 227).

2.3 DSGE estimation via IRF matching

Theoretical models are useful to economists and policymakers only if they are able to reproduce important features of the observed data. Which is a property that critically depends on the values that model's parameters have attached, and also on how the researcher decide to estimate them Theodoridis (2011).

Altig et al. (2005) following Rotemberg and Woodford (1997); Christiano et al. (2005); Iacoviello (2005); Boivin and Giannoni (2006), use a limited information approach, an IRF matching estimator, this procedure select and estimate the model's structural parameters that produce the DSGE IRFs, which minimises the distance with IRFs created from the SVAR.

Using an IRFME implies that the researcher is focusing to match on some limited aspects of the DSGE Giacomini (2013); it is choosing an identification strategy for the reduced form VAR in a way that is consistent with the key determinants of the DSGE. The VAR and theoretical IRFs are identified by restrictions implied by the DSGE. Which means that the DSGE admits a structural VAR representation, in this way, sample moments of the VAR are informative for the theoretical model's structural parameters Rossi et al. (2012).

Theodoridis (2011), following Canova (1994, 2007), notes that many DSGE economists prefer to use partial information estimators rather than full information ones, like maximum likelihood, because they prefer estimates that minimise the economic distance rather than statistical fit criteria. Because structural IRF matching estimates deliver selected DSGE moments-statistic as close as possible to those observed in the data. The economist can assess how well the estimated model replicates these targets allowing the researcher to identify possible modelling weaknesses and to infer useful economic conclusions.

The authors use a medium-scale VAR with four lags, ten quarterly series and three structural shocks. The monetary policy shock is identified with a short-run recursive Cholesky restriction following Christiano et al. (2005). The two technology shocks, neutral and capital embodied, are identified instead with long-run restrictions, following Fisher (2006), which builds on Shapiro and Watson (1988). As summarised by Rossi et al. (2012), the restriction imposed by Altig et al. (2011) is that the two technology shocks are the only shocks allowed to affect the level of labour productivity in the long run; among them, only the capital embodied shock is allowed to affect the relative price of investment in the long run. The VAR is identified so that its impulse responses satisfy these restrictions, and the resulting responses are then matched to their model counterparts to estimate the structural parameters.

Although the model is estimated from the responses to three shocks, the question that motivates Altig et al. (2011) is a monetary one. What the paper sets out to explain is why inflation responds so slowly to a monetary policy shock while firms in the micro data change prices often, and the resolution runs entirely through χ, which is governed by the cost of varying capital utilisation.

That cost is a statement about what happens to a firm's marginal cost when demand for its output moves while its capital stock is fixed, and a monetary policy shock is exactly

that experiment. It shifts demand within the quarter, technology is unchanged, and the capital stock is predetermined, so the joint response of utilisation, hours and inflation is close to a direct reading of Capacity utilisation cost σ_a.

Capacity utilisation is itself one of the ten series in the system, so this is not an indirect inference. A technology shock is a poorer experiment for the same parameter, because it moves the production function and the desired quantity of capital services at once, and what the utilisation response reveals is then a mixture of the two.

This is not to say that the technology shocks are decorative. The DSGE carries three structural disturbances and the estimator uses all of them: they carry information about the investment adjustment cost and about the parameters governing the propagation of supply-side disturbances, and without them several parameters would be poorly pinned down. But they are not where the micro–macro conflict is settled. It is settled by σ_a, and σ_a is read off the response to a monetary policy shock.

The analysis that follows in this work therefore centres on the monetary policy shock. If σ_a turns out to be sensitive to the VAR's specification, so is the reconciliation Altig et al. (2011) propose, and the sensitivity comes from the monetary policy block, the part of the model this paper is really about.

The two technology shocks are nonetheless reported and discussed throughout, rather than set aside. They are identified in a different way, through restrictions on their effects in the long run rather than on their behaviour within the quarter, and that difference will prove informative in its own right.

2.4 SVAR and IRF minimum distance

The SVAR can be recovered by imposing structure on the reduced form (RF),

$$B_0^{-1}B_0Y_t = B_0^{-1}B_1Y_{t-1} + \cdots + B_0^{-1}B_pY_{t-p} + B_0^{-1}w_t, \tag{3}$$

which matches the RF-VAR term by term once $A_i = B_0^{-1}B_i$ for each lag i and

$$u_t = B_0^{-1}w_t. \tag{4}$$

Equation (4) is the bridge between the two representations: the reduced-form residuals are a linear combination of the structural shocks. Taking variances on both sides,

$$\Sigma_u = B_0^{-1}\Sigma_w\big(B_0^{-1}\big)'. \tag{5}$$

Σ_w is the variance-covariance matrix of the structural shocks w_t. These are assumed mutually orthogonal and normalised to unit variance, so $\Sigma_w = I_K$ by construction. This is

a normalisation, not an assumption about the economy: what is estimated is not the scale of each shock, but how a given shock propagates through the system, which is entirely encoded in B_0^{-1}.

Y_t is the ten-dimensional vector of series used by Altig et al. (2011),

$$Y_t = \begin{pmatrix} \Delta \ln(\text{relative price of investment}_t) \\ \Delta \ln(\text{GDP}_t/\text{Hours}_t) \\ \Delta \ln(\text{GDP deflator}_t) \\ \text{Capacity utilisation}_t \\ \ln(\text{Hours}_t) \\ \ln(\text{GDP}_t/\text{Hours}_t) - \ln(W_t/P_t) \\ \ln(C_t/\text{GDP}_t) \\ \ln(I_t/\text{GDP}_t) \\ \text{Federal funds rate}_t \\ \ln(\text{GDP deflator}_t) + \ln(\text{GDP}_t) - \ln(\text{MZM}_t) \end{pmatrix}. \tag{6}$$

All the variables are quarterly series from the 1959Q2 - 2001Q4 period, more details on data are described in Appendix A.

Identification combines a short-run and a long-run restriction. The full derivation is in Appendix B.

These restrictions identify three shocks out of ten. IRFME matches the IRFs of the three identified shocks, not on those not identified.

2.5 Estimating the Parameters of the Equilibrium Model

Using Altig et al. (2011) notation, let ζ denote the structural model parameters that are aimed to be estimated, and $\Psi(\zeta)$ being the mapping from the structural parameters ζ to the model IRFs. The SVAR IRFs estimates are denoted as $\hat{\Psi}$.

The IRFME then can be written as:

$$\hat{\zeta} = \arg\min_{\zeta} (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta)) \tag{7}$$

The IRF matching is done horizon by horizon: the estimator compares the VAR point estimate of each impulse response at each of the first h horizons; being $h = 20$ in Altig et al. (2011) work, excluding those set to zero by identification, against the model-implied counterpart. Here, V is a diagonal matrix composed by the sample variances along the diagonal $\hat{\Psi}$. Those variances, compose the confidence intervals around the point estimates of the SVAR IRFs. With this choice of V, the estimated model's structural parameters $\hat{\zeta}$ are the value of ζ that ensures that the $\hat{\Psi}$ DSGE's IRFs lies as much as possible inside the VAR IRFs' confidence intervals. Standard errors of the DSGE's structural parameters

$\hat{\zeta}$ are computed using the Delta method, more detail in the Appendix G

The minimum distance estimator relies on the ability of the VAR to estimate the IRFs. *"if the VAR is misspecified, the specification errors will be computed at each horizon"* point the attention on misspecification concerns in general VAR practice (Ramey, 2016, p. 83). This is clear from the authors themselves, who claim *"The quality of our strategy (IRF matching) depends on the ability of identified VARs to generate reliable estimates of the dynamic response of economic variables to shocks"* (Altig et al., 2011, p. 3, fn. 10).

3 Truncation Bias and the Case for Long-Lag VARs

Many DSGE's log linearised solution can be mapped in State Space representation, and then, under some assumption can be re-written as VAR(∞) Fernández-Villaverde et al. (2007). Standard lag choice by SVAR practitioner is to use forecasting information criteria (IC) such as Akaike (AIC) or Hannan-Quinn (HQ) and Schwartz (SW), but the most common choice is use conventions; that is, 4 lags for quarterly data and 12 for monthly data.

If a DSGE is estimated using an IRF matching technique, using a short lag VAR it might be problematic as the DSGE is estimated by a biased by lag truncation VAR. Truncation Bias is a type of misspecification that arises when the VAR lag order is shorter than that of its DGP. This type of misspecification might be a potential problem for structural inference.

VARs are typically small systems with few lags often interpreted as models of the entire economy, IRFs are highly sensitive to variable and lag choice Barnichon et al. (2026); De Graeve and Westermark (2025).

3.1 Illustration of the problem: Truncation Bias

Following the example used by De Graeve and Westermark (2025), consider a data generating process in which y_t depends on two exogenous variables,

$$y_t = x_{1t}\beta_1 + x_{2t}\beta_2 + \epsilon_t. \tag{8}$$

Estimating the relationship while omitting x_{2t},

$$y_t = x_{1t}b_1 + e_t, \tag{9}$$

Equation 9 delivers a biased point estimate unless the regressors are orthogonal, that is $\mathbb{E}(b_1) \neq \beta_1$, and an upward-biased residual variance, $s^2 > \sigma^2$. Both distortions matter here, and the second is the one usually forgotten.

The same logic carries over to a VAR. Let

$$Y_t = A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \tag{10}$$

with

$$A(L) = A_1 L + \cdots + A_p L^p \tag{11}$$

where $p \ll p^*$, where p^* denotes the true lag order.

Truncation biases both the reduced-form autoregressive coefficients $A(L)$ and the reduced-form covariance matrix Σ_u. Since impulse responses are a function of both, they inherit the bias.

$$\text{IRF} = f\big(A(L), \Sigma_u\big), \tag{12}$$

Writing the impulse responses this way suggests that Σ_u enters structural analysis on its own. It does not: Σ_u is only ever used through a further mapping into the structural shocks, via the impact matrix B_0, and whether that mapping inherits the truncation bias just described depends entirely on how the shock of interest is identified.

3.2 Identification Bias

Ravenna (2007) finds two channels through which lag truncation operates. The first being *pure truncation* bias the one described in the section above. The second he calls it *identification bias*, and works through the fact that impulse responses are not read off the reduced form alone, they also require the map from reduced form innovations to structural shocks, and that map has to be estimated as well.

Ravenna's identification bias arises when the identifying restrictions are correct, so it does not originate in mistaken identification assumptions, and it arises in population, so it is not an artefact of small samples.

Whenever the estimated autoregressive coefficients enter the identifying map's computation, a biased $\hat{B}(L)$ delivers a biased $\hat{B}_0$, and the object the researcher labels a structural shock is not the shock the identifying assumption defines.

Whether the second channel is present, therefore reduces to a single question: do the estimated autoregressive coefficients enter the computation of the identifying map?

Under a recursive short run scheme they do not. The identifying restrictions are zero restrictions on contemporaneous relations, so B_0 is recovered from $\hat{\Sigma}_u$ alone, and the lag polynomial plays no part in it. This does not make the recursive scheme immune to pure truncation, since $\hat{\Sigma}_u$ is itself inflated by the omitted dynamics and the propagation is distorted as it is for everyone else. The short run Cholesky scheme is not affected to

the second channel, identification bias, because the restriction is a statement about the contemporaneous covariance of whatever innovations the estimated VAR produces, and it retains that meaning at any lag order.

Under a long run scheme they do. Long run identification is known to be delicate for several reasons, of which truncation is only one Faust and Leeper (1997); Erceg et al. (2005); Chari et al. (2008); Christiano et al. (2007). The mechanism at work here is clearest in the formulation of Blanchard and Quah (1989), where the identifying map is obtained by factorising the long run covariance matrix

$$[I - B(1)]^{-1} \Sigma_u \left\{[I - B(1)]^{-1}\right\}', \qquad B(1) = \sum_{j=1}^{p} B_j, \tag{13}$$

so that the entire lag polynomial evaluated at unity enters the identification directly. A VAR(p) does not estimate that sum consistently when the true representation is of infinite order, and the restriction is therefore imposed on the wrong object.

Altig et al. (2011) impose the same kind of restriction by a different route, following Fisher (2006): instead of rotating an estimated matrix after the fact, they build the restriction into the estimating equation and read the identified shock off as a regression residual (mechanics in Appendix B). Either way, the restriction binds on the sum of the lag polynomial the VAR actually estimates, which runs only to p, not on the sum over the true $p^* > p$ lags it is meant to capture. The restriction is anchored to the wrong object, and the shock it delivers is the wrong linear combination of reduced form innovations.

This has a consequence for where the bias may appear. A long run restriction says nothing about any finite horizon, impact included. It fixes the cumulated effect and leaves every response along the way to be implied by the estimated dynamics. The long run restriction is therefore not a constraint on the response at a given horizon: it is the rule by which the shock is isolated from the reduced form innovations, and it is on that rule that truncation acts. Applied to a truncated polynomial, the identifying restriction can isolate a different combination from the intended one. If the shock itself turns out to be a different one, the bias is no longer bound to accumulate with the horizon: it could already be present at the first period, unlike pure truncation bias which builds up over time.

The two channels leave different traces: *pure truncation* bias distorts propagation, so its mark accumulates with the horizon, while *identification* bias distorts the shock, so its trace could be visible already on impact.

The VAR of Altig et al. (2011) carries both kinds of restriction at once, which makes it an unusually clean setting in which to look for both the channels.

In essence, both the channels in which truncation bias potentially operates can be closed or mitigated by simply increasing the lag length of the VAR. Applied practice has largely resisted that response. Even where long-lag VARs are known to have good asymptotic properties Plagborg-Møller and Wolf (2021), they are widely treated as unworkable in finite samples due to high variance concerns: more lags mean more parameters, fewer degrees of freedom, and confidence intervals presumed to widen accordingly. That presumption is part of why a recent literature prefers Local Projections (LP) to VARs outright Li et al. (2024); Montiel-Olea et al. (2025).

De Graeve and Westermark (2025) explore the trivial solution to short lag VARs, that is, dramatic increase in the lag length.

3.3 Long-Lag VARs

Their central point is that standard intuition that, increasing lag length increase uncertainty around estimates given the increased parametrisation is just a portion of the story, *"in face of misspecification due to lag truncation, increasing the lag length can actually reduce uncertainty"* (De Graeve and Westermark, 2025, p. 2).

As the dynamic structure increases, lag truncation decrease, consequently misspecification reduces. Misspecification reduction has a double effect, the first being the well-known bias reduction, while the second is the variance reduction that works against the imprecision coming from the increased parametrisation. This trade off it applies to all truncated VARs, no matter what the identification strategy is.

To reveal Long-Lag VARs properties De Graeve and Westermark (2025) generate data, with a Monte Carlo, of realistic samples size from three DSGE. The three DSGE are Chari et al. (2008), Christiano et al. (2007) and Smets and Wouters (2007).

This Monte Carlo simulation is conducted under the assumptions that the identification restriction is correct. And the DSGE are fundamental in the sense of Fernández-Villaverde et al. (2007), so invertibility is never a problem. Under these hypothesis these models have an exact VAR(∞) representation.

With the Monte Carlo they generate 1000 datasets of realistic macro sample size for quarterly data, $T = 200$, and they estimate the bivariate VARs [2] representations of the DSGE choosing different increasing the lag choice. When comparing VARs of different lag-length, they restrict the sample of effective observations as $T = 170$ ensuring that lag initialisation does not affect sample size. Uncertainty bands are computed as in (Sims

[2]For Smets and Wouters (2007), they reduce the number of variables from 7 to 2 to maintain consistency with the other two exercises. They keep GDP-growth and interest rates.

and Zha, 1999; Canova, 2007; Uhlig, 2005), in this sense it is used a the Normal-Inverse Wishart prior, which have good approximation to frequentist bootstrapped intervals. It does not weight differently short vs. long lags as would do other priors like the Minnesota one.

3.4 Long-Lag Benefits

De Graeve and Westermark (2025) analyse how the truncated VARs of the three DSGE with VAR(∞) representation behave. The results are much in favor for long-lag VARs. The variance-reducing effect of lower misspecification often dominates the penalty of increased parametrisation. In a nutshell, Long-Lag VARs, reduce truncation bias, have more precise inference, reduced Mean Squared Error (MSE) and better coverage.

Short-lag VARs have maximum uncertainty for horizons where uncertainty is not mechanically low. For longer horizons, short-lag VARs have minimum CI width, this because a VAR(p) cannot propagate much beyond horizon p, VARs with lag length extrapolate impulse responses for horizons $h > p$ from the first p autocovariances Baumeister (2025). Therefore uncertainty cannot propagate much beyond that horizon either.

Uncertainty does not explode with the lag length increase; instead for VARs with very long lags, confidence bands are roughly in the same ballpark as those of short-lag VARs.

Long-Lag VARs have comparable coverage rates and comparable MSE than short-lag VARs, combined with the tendency to produce smaller biases, long-lag VARs have favorable properties compared to standard short-lag VARs.

De Graeve and Westermark (2025) leverage to the above mentioned Monte Carlo simulation proposing an exercise that decompose the uncertainty effects creating hypothetical IRFs that are composed by long-lag estimates of the point estimates, while using short-lag estimates of variance covariance matrix obtaining an $IRF = f(B_{30}(L), \Sigma_4)$. Such IRFs can be interpreted as isolating the effect of increased parametrisation.

The standard intuition would suggest that long-lag length makes the entire distribution shift outward, through the additional uncertainty created by the strong increase of parameters. In their exercise, they find that the increased parametrisation does not invariably increase uncertainty, which is at odds both with standard intuition (fewer degrees of freedom) and with the well known omitted variable result.

Standard intuition on increased parametrisation is partially correct and obviously part of the story, but De Graeve and Westermark (2025) finds that misspecification reduction tends to have substantial variance effects.

3.5 Choosing the Lag Length

Macroeconomists almost never have the luxury of knowing the true DGP, unlike in the Monte Carlo exercise just reviewed, so point estimates cannot be checked against a known

truth the way De Graeve and Westermark (2025) do in simulation.

What can be checked instead is robustness: whether the structural conclusions hold up as the lag length increases substantially, or shift as p grows. Stability across lag orders is the closest a researcher can come to the reassurance the Monte Carlo evidence provides, which is why Baumeister (2025), and De Graeve and Westermark (2025) themselves both ask whether long-lag VARs remain a viable methodology in sample data, revisiting well known empirical studies rather than relying on simulation alone: De Graeve and Westermark (2025) return to Galí (1999), where the evidence against RBC models in favour of New Keynesian ones weakens with longer lags; to Chang and Hong (2006), whose central result is similarly less strong; and to Miranda-Agrippino and Ricco (2021), where longer lags change the estimated duration of the monetary policy shock's impact.

This robustness check is something a researcher cannot choose to skip, for two reasons. The first applies whenever the economic question is itself about the long run: a VAR(p) cannot propagate a response, or say anything informative, beyond horizon p, so a long-run question mechanically requires a long-lag VAR. Baumeister (2025) illustrate the point directly: re-estimating the BVAR of Jordà et al. (2024) with 60 monthly lags eliminate its conclusions on the long-run effects of monetary policy. Antolín-Díaz and Surico (2025) build the same requirement into their design from the outset, using $p = h$ lags to draw structural inference at horizon h.

The second feature applies even when the question is not about the long run at all, and it is the point I want to stress here. When the DSGE is estimated by IRF matching, the estimator does not target a single horizon: it minimises the distance between VAR and DSGE impulse responses jointly over h periods, up to $h = 20$ in Altig et al. (2011). If the VAR misspecifies the dynamics it is asked to estimate, that misspecification is present at every one of those horizons, and the estimator is matched to an object that is biased throughout the window, not only near its end. Antolín-Díaz and Surico (2025) set $p = h$ because their question is explicitly about the long run; what this thesis adds is that under IRF matching the same requirement follows even when it is not, because the estimator's own objective function reaches out to horizon h regardless of what the underlying economic question asks about. A researcher whose question is genuinely about short-run dynamics is therefore not exempt from the choice of p: as long as the estimation strategy is IRF matching, p has to be chosen with h in mind, not only with the horizon of the substantive question.

The mismatch can equally be resolved from the other side, by lowering h: twenty quarters, five years, is a long horizon for a question that is meant to be about the short run, and a shorter matching window asks less of the VAR's dynamics. Nor need the

two adjustments be alternatives; raising p and lowering h together, so that neither lever does all the work on its own, converges naturally on $p = h$, which is exactly the choice Antolín-Díaz and Surico (2025) make.

None of this is an argument for going long everywhere. De Graeve and Westermark (2025) are explicit that their results speak to structural inference, not to forecasting: when the objective is predicting unseen data, parsimony still earns its keep, because overfitting rather than misspecification is the main risk there. It is only when the objective is structural, be it impulse responses, causal inference, or IRF matching against a DSGE, that the misspecification concerns described above become the ones that matter.

This is also why the conventional criteria used to choose p, Akaike, Hannan-Quinn and Schwarz, are the wrong tool for that objective. They are built to minimise out-of-sample forecast error, which rewards parsimony, not to protect against the truncation bias Barnichon et al. (2026); Baumeister (2025). Both Baumeister (2025) and De Graeve and Westermark (2025) argue that a researcher whose aim is structural inference should not defer to these forecasting criteria, and González-Casasús and Schorfheide (2025) propose an alternative designed for structural inference.

There is no need, however, to wait for that or other criterias to become standard practice. The Long-Lag VARs authors already offer a simpler heuristic that any researcher can apply today: *"plot the structural moments central to the research question against lag length. If the structural conclusion dramatically change as the lag length increases, that is clearly indicating that truncation is important to the question at hand"* (De Graeve and Westermark, 2025, p. 12).

3.6 Model size

The bias-variance trade-off documented by De Graeve and Westermark (2025) applies to all untruncated reduced form VARs. It does not depend on the number variables in the VAR, nor in the identifying assumption or whether the researcher is using a Bayesian or Frequentist method.

The Frequentist researcher might bump into a degrees of freedom problem. An n-variable VAR(p) has $n + n^2p$ coefficients, and $\frac{n(n+1)}{2}$ elements in the variance covariance matrix. As the lag length grows, the number of parameters to estimate will rapidly increase, and the higher is the number of variables in the VAR, the faster will happen. At some point the number of parameters to estimate will be greater than the number of available data points. It is shown by De Graeve and Westermark (2025) that the degrees of freedom limit is far from being reached by small scale VAR with one year of lags. In

this work replicating Altig et al. (2005) we stand in the Frequentist side, so I won't talk regularization techniques or how easy it is to implement long-lag VARs in Bayesian as they inherently avoid over-parametrisation.

Given the dimension of VAR in analysis, I am going to show that even being close to the limit, it is possible to estimate long(er)-lag VARs in the Frequentist approach.

4 Empirical Results

In this section I present results from enriched dynamic structure of Altig et al. (2005) VAR used to estimate their DSGE parameters. Doing this exercise, I extend the work of De Graeve and Westermark (2025) in three different aspects.

First, in Monte Carlo laboratory that De Graeve and Westermark (2025) have created to evaluate the three DSGE that have chosen to use to generate data, all respect an important assumption, Fernández-Villaverde et al. (2007) Poor Man's condition. All the DSGE admits a VAR(∞) representation, and because of this they can evaluate the performance of truncated VARs, which are fully informative. They are bivariate, with two variables and two shocks. The VAR used in Altig et al. (2011) confronts three shocks with ten observable variables. The system is therefore stochastically singular, there is no exact VAR(∞) representation of the DSGE that the VAR is being used to estimate for. But this is not a caveat, is a feature, and is one of the reasons why the Altig et al. (2011) used to estimate their model a partial information approach as IRF minimum distance estimator. In addition the "fundamentalness" concept in the Lippi and Reichlin (1993); Fernández-Villaverde et al. (2007) sense, is extrimely and unnecessarly restrictive, and in several cases non needed as stressed by Forni et al. (2016), as the researcher is often interest in assessing the transmision mechanism of a single or few shock.

The second departure as previously anticipated in the Section 3.3 relates on the dimension of the VAR employed in this work. Because the three DGP processes in De Graeve and Westermark (2025) are two-variable systems, their VAR remains cheaply parametrised even at long lag lengths, and increasing p costs relatively little in terms of degrees of freedom. The ACEL VAR does not enjoy that luxury. With $K = 10$ variables, a constant and p lags, each equation carries $10p + 1$ coefficients, and the sample runs from 1959:Q2 to 2001:Q4, that is 171 quarterly observations before lags are consumed. I will show that concerns related to CI explosions might not occur even in limiting cases.

Third, which is strictly connected to the second, I am using as baseline result the exercises where I am not restricting the sample size to the one used in the longest-lag VAR. The reason is why the results does not vary much, and when they do they do by a small amount and consistently with the baseline when the common available data is

used. It is shown in Appendix I. The other reason is that, in practice, if a researcher has to decide to use a longer-lag VAR rather than a short one, the short VAR would be still estimated with as many data points available. In particular this argument is even more valid when the VAR has a lot of variables or if the available data points is little.

4.1 VARs IRFs

The baseline lag choice of ACEL's VAR is $p = 4$, I re-estimate their DSGE matching it with VARs of lag $p = 8$ and 12, raising the lags to one up to two and three years. In Appendix I are presented the results when the sample is restricted for all the estimations to the one available to the VAR with twelve lags.

In Figure 2 are plotted the VARs' IRFs to a one standard deviation monetary policy shock easing across different lag structures with 95% bootstrap confidence intervals of the VAR(4). All variables, except for inflation and the interest rate are expressed in percent terms. Inflation and Fed Funds Rate are expressed in units of percentage points at quarterly rate and annual rate, respectively.

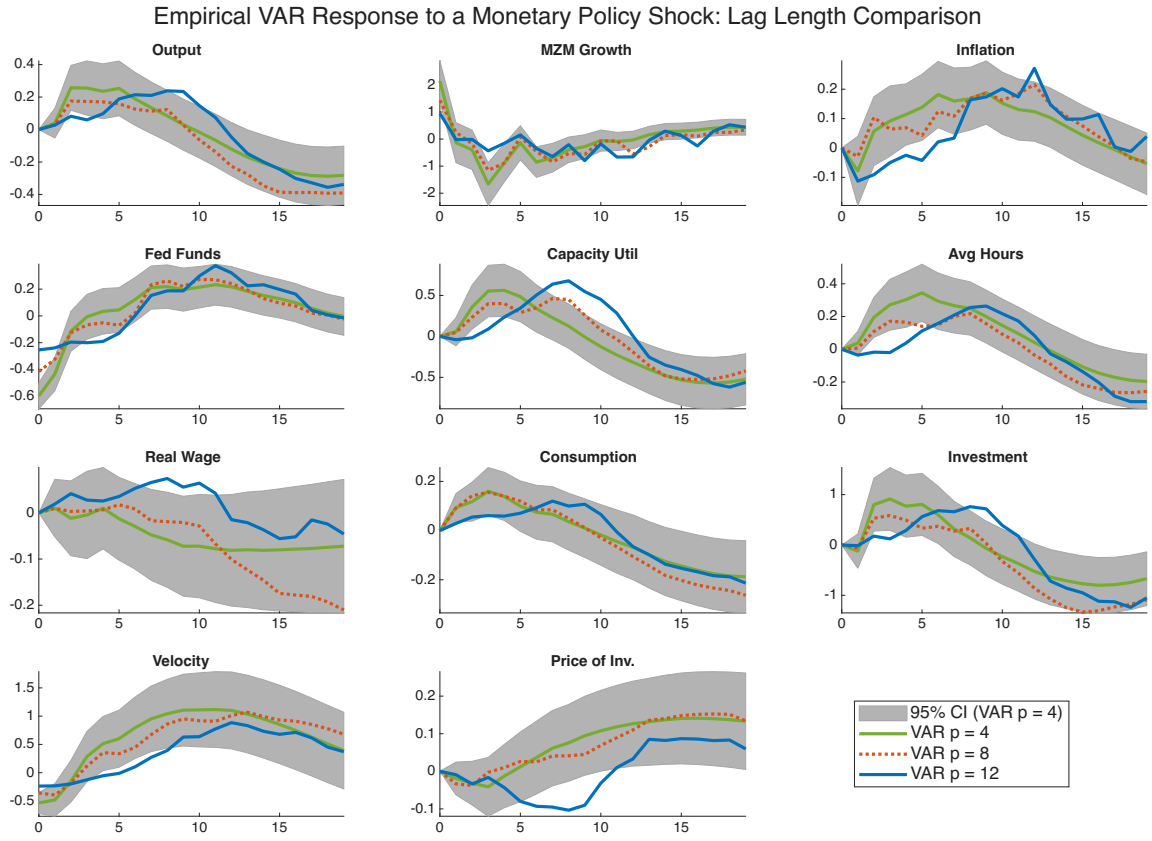


Figure 2: VARs' IRFs to a Monetary Policy shock across different lag structure

The green solid line with grey 95% bootstrap confidence interval are shown the Altig et al. (2005) baseline results from the VAR with four lags. In orange dashed line the IRFs

VAR(8), and in blue solid line the VAR(12).

I start describing Figure 2 from what does not change dramatically. First, the response of money growth rate to a monetary policy shock, consistently with the baseline VAR(4), the effect is completed within one year. Second, there is a liquidity effect, money growth and interest rate move in opposite directions after the shock. Third, velocity co-moves with the interest rate, which falls in response to the shock and then rises. Fourth, the price of investment seems not to be affected on impact, though it shows some divergence from the baseline at intermediate horizons. In general, all the point estimates across different lags lie inside the confidence intervals of the VAR(4).

The real wage in the VAR(12) has a hump shape, compared with the flat baseline.

Some important responses do not change in magnitude, sign or shape, but only in timing. For output, inflation, capacity utilisation, average hours, consumption, velocity and investment, as the lag choice increases, the timing of the hump's formation and its peak shifts later. On average, the peak effect in the longer-lag VARs occurs about one year later than in the baseline results, with the exception of consumption in the VAR(8), whose response is indistinguishable from the baseline.

The size of the Fed Funds rate shock changes from 60 basis points at horizon $h = 0$, decreasing almost to 0 within two quarters, to a more realistic 25 basis points shock that lasts for about five quarters before going to 0.

The VAR(12) inflation response needs some more attention. Altig et al. (2005) themselves report that, in their own baseline VAR, inflation falls for a single quarter after the easing shock before rising to its peak roughly two years later. At $p = 12$ that brief dip turns into a price puzzle: inflation stays negative for the whole first year, and only starts rising from the second year onward, reaching its peak around the third year rather than the second. It is much more inertial than the baseline, but that is consistent with every other variable in the VAR(12) also being more inertial. Having in mind Altig et al. (2005)'s theoretical mechanism, the delay is itself informative: with such strong nominal frictions, only a small fraction of firms reset their price in any given quarter, so the shock takes longer than a year to work its way through inflation, well after it has already worked through output, consumption and average hours.

Given the persistence and slow response of the variables in the VAR(12), this is consistent with the Altig et al. (2011) mechanism only if price stickiness is high, since high stickiness is exactly what implies a high implied duration of prices, that is, firms holding their prices fixed for a long time between resets.

4.2 Technology shocks and the long run restriction

The two technology shocks are identified with long run restrictions following Fisher (2006), so this is where the second channel found by Ravenna (2007), the identification bias

described in Section 3.2, could bind.

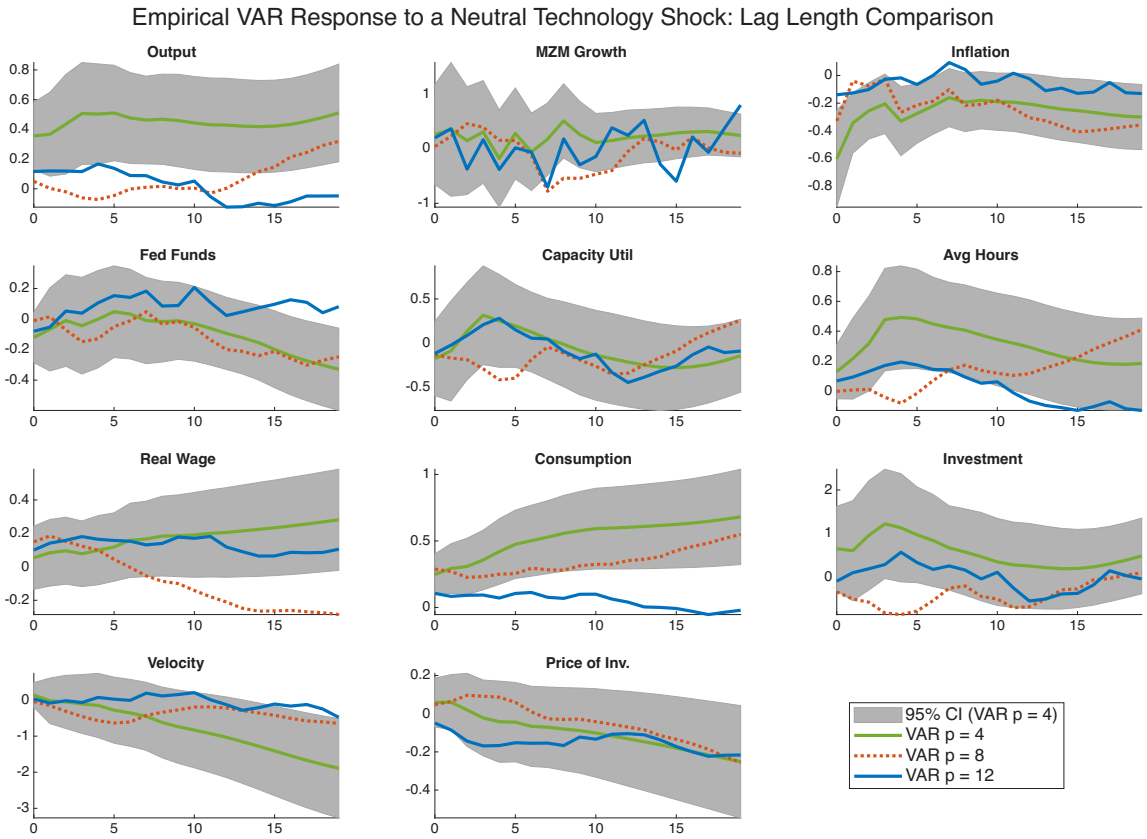


Figure 3: VARs' IRFs to a Neutral Technology shock across different lag structure

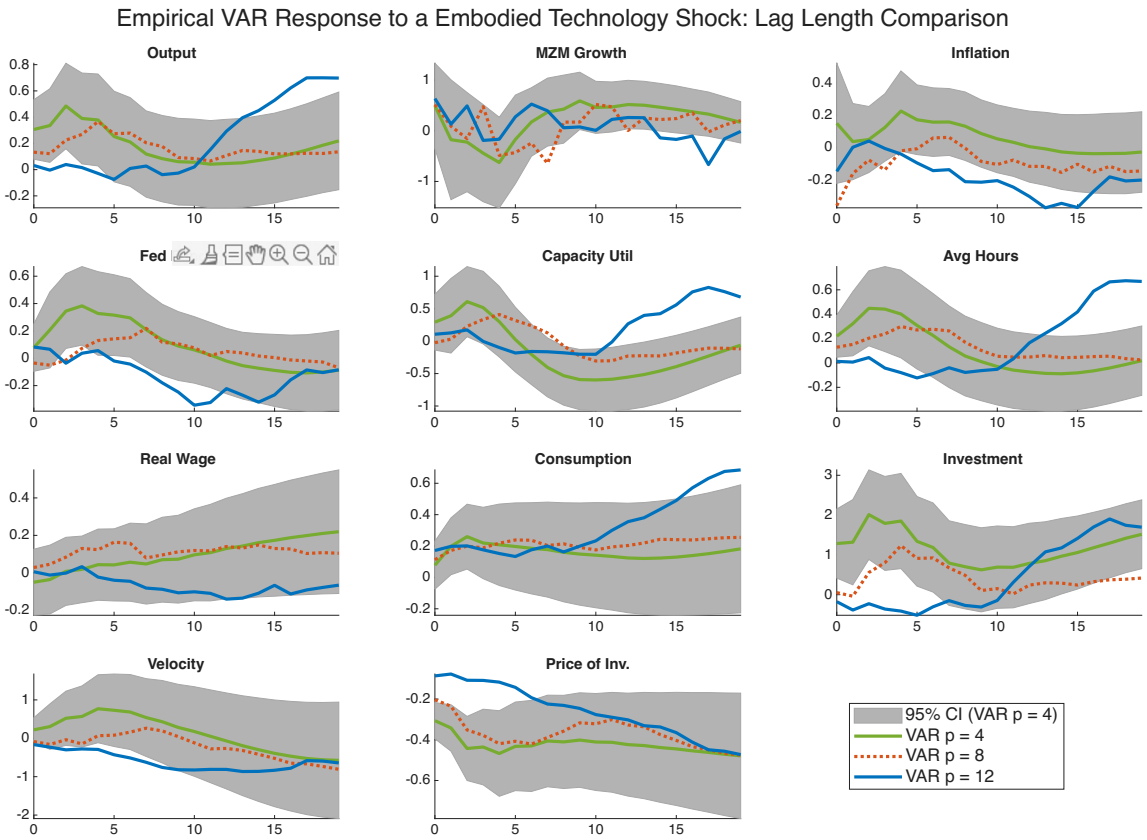


Figure 4: VARs' IRFs to a Embodied Technology shock across different lag structure

Figure 3 reports the responses to a neutral technology shock, and the disagreement across lag orders is of a different nature from the one found for monetary policy. It is not a matter of timing, the responses do not place the same hump later, they change level and, for several variables, sign. Output rises by around half a percent and stays there in the baseline VAR(4), while in the VAR(12) it starts close to zero and drifts below it after the third year. Consumption behaves in the same way, positive and persistent at $p = 4$, flat and indistinguishable from zero at $p = 12$. Average hours are the clearest case, the baseline delivers a positive hump of about half a percent, the VAR(12) a flat response that turns negative at intermediate horizons. The sign of the hours response to a neutral technology shock is the object of a long debate (Galí, 1999; Christiano et al., 2007), and here the lag order alone settles it in either direction.

Figure 4 shows the same instability in a different form. The baseline responses to an embodied technology shock are front loaded: output, average hours, capacity utilisation and investment all peak within the first four quarters and then decay. The VAR(12) responses are close to zero, or slightly negative, over exactly that range, and they rise only after the tenth quarter, reaching their maximum at the end of the horizon. The profile is not delayed, it is reversed: where the baseline sees the effect of the shock exhausting itself, the long lag VAR sees it beginning.

What matters for the argument is what happens on impact. Under the recursive

scheme most impact responses to a monetary policy shock are zero by construction, and those that are not differ across lag orders roughly in proportion to the estimated size of the shock. Nothing of the sort holds here. The impact response of the relative price of investment to an embodied technology shock, the variable on which the restriction is anchored, falls from about -0.45 in the baseline to about -0.1 at $p = 12$. Inflation responds positively on impact at $p = 4$ and negatively at $p = 8$. Investment responds positively on impact to a neutral technology shock at $p = 4$ and negatively at both longer lag orders. These are disagreements at horizon zero, before any propagation has taken place.

This is the trace that Section 3.2 attributes to identification bias rather than to pure truncation. Pure truncation distorts the propagation of a correctly isolated shock, so its mark has to accumulate with the horizon and cannot by itself generate a disagreement at $h = 0$. A restriction imposed on $B(1)$ computed over p lags, when the true polynomial runs to $p^* > p$, isolates a different linear combination of the reduced form innovations, and a different shock has a different impact effect. The monetary policy responses, identified without the lag polynomial, keep their sign and their shape and move only in timing. The technology responses, identified through it, do neither. This is the asymmetry Ravenna (2007) predicts, and it appears here within the same VAR, on the same sample, with only the identification scheme differing.

Two qualifications are in order. The two channels are not mutually exclusive and the technology responses are subject to both, so what the figures establish is that the long run identified responses are the fragile ones, not that pure truncation is absent from them. And longer lags do not by themselves certify the responses at $p = 12$ as the correct ones. The claim is the weaker one: what a technology shock does in this VAR is not robust to the lag order, and it is fragile in exactly the place where the theory says it should be, which is also the place Chari et al. (2008) target in their critique of long run identified SVARs.

4.3 Uncertainty around longer lags estimates

Conventional view thinks that increasing lag length increase parametrisation, therefore reduce the degrees of freedom making uncertainty around the estimates larger. As well explained by De Graeve and Westermark (2025), this is for sure part of the story. But the long-lag VARs authors demonstrate how misspecification lowers the variance of the residuals, and that can offset the imprecision caused by additional parameters. As previously mentioned, in a bivariate system there is plenty of room to increase lag length without having big concerns relative to overparametrisation. In Altig et al. (2011) VAR the concerns are more than realistic, the parametrisation grows by a factor of three already large relative to the sample. If confidence intervals were going to explode anywhere, they

would do it here.

They do not. Table 1 summarise the coefficients and degrees of freedom of the exercise. The baseline exercise at $p = 4$, the VAR has 410 coefficients, 167 available data points and 41 coefficient per regression. At $p = 12$ the system carries 1210 coefficients, each equation is estimated on 159 observations with 121 regressors, and only 38 degrees of freedom survive. The degrees of freedom limit, at which the number of regressors per equation k is greater than the number of usable observations T_u, pointed out by De Graeve and Westermark (2025) in this exercise is at $p = 16$.

Tripling the lag length relative to the ACEL baseline is therefore not a conservative perturbation, but it is the largest perturbation the data permit. This makes the exercise a demanding test for the central claim in De Graeve and Westermark (2025).

Table 1: Degrees of freedom across lag orders

	VAR $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Usable observations T_u	167	163	159
Coefficients per equation k	41	81	121
Degrees of freedom $T_u - k$	126	82	38
Observations per parameter T_u/k	4.07	2.01	1.31
Coefficients, full system	410	810	1,210

Note: T denotes the raw sample size and T_u the usable sample size, $T_u = T - p$; k denotes the number of regressors per equation, $k = 10p + 1$ ($K = 10$ variables, p lags, one constant). The raw sample contains $T = 171$ quarterly observations (1959:Q2–2001:Q4), of which p are consumed as initial conditions, so that $T_u = T - p$. Ordinary least squares requires $k < T_u$, that is $10p + 1 < 171 - p$, that is $p \leq 15$; at $p = 15$ only 5 degrees of freedom per equation remain.

Figure 5 plots the VARs' IRFs to a monetary policy shock at each lag order, together with their own 95 percent bootstrap confidence bands.

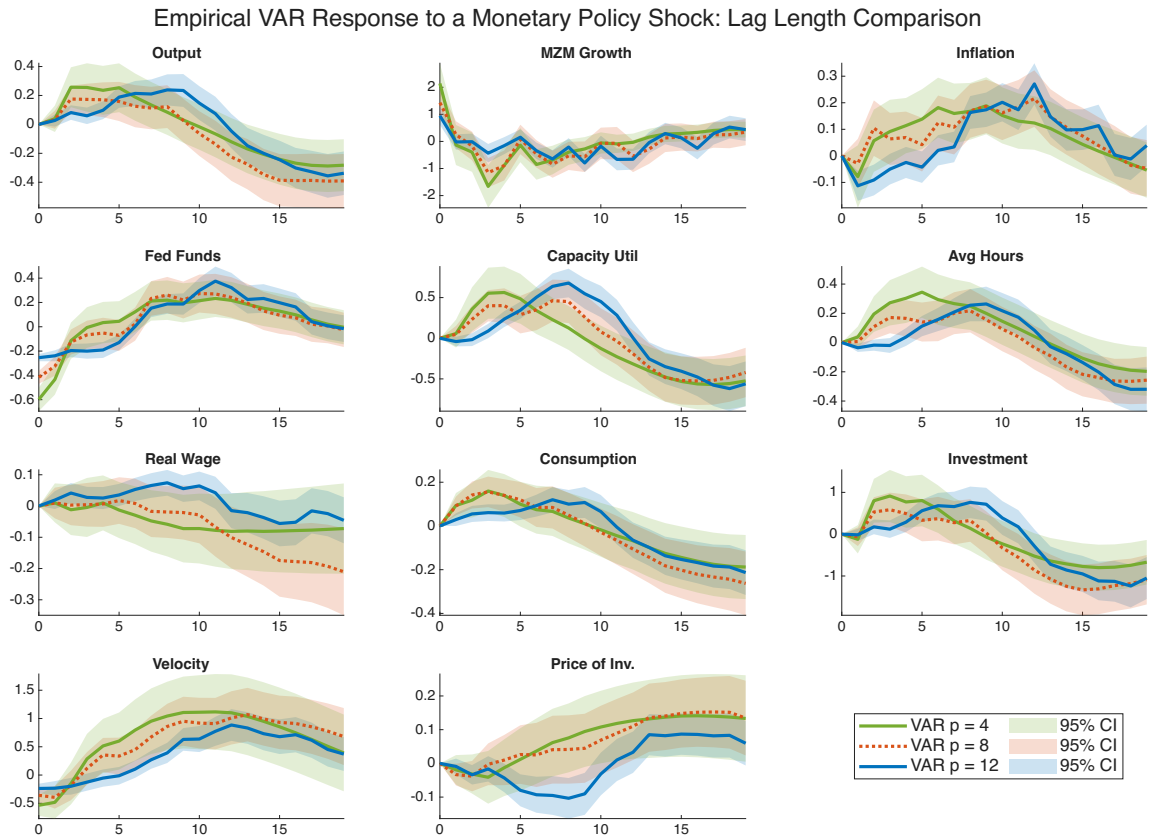


Figure 5: VARs' IRFs to a Monetary Policy shock across different lag structure, 95% CI

The bands at $p = 12$ are not wider than those at $p = 4$ for several variables; inflation, Fed Funds rate, real wage, velocity and relative price of investment, they are visibly narrower despite the increased parametrisation documented in Table 1. The pattern is not confined to those five variables. The bands at $p = 12$ are narrower than at $p = 4$ at every horizon and for every variable in the system, with the single exception of MZM growth, which is the only variable whose bands widen at some horizons. Therefore suggesting that the variance reduction mechanism of De Graeve and Westermark (2025) operates even at the parametrisation frontier, at least in this work.

Is this just overfitting?

The first objection would be that this is simply overfitting: a VAR with 121 regressors per equation fits the sample closely, the estimated residuals shrink, and confidence bands computed from those residuals shrink with them. The objection needs to be taken seriously, because in sample it cannot be dismissed; reduced misspecification and overfitting have the same in sample signature, namely a smaller residual variance.

What the objection does not deliver, however, is the conclusion it appears to deliver. Overfitting pushes the width of the bands in two opposite directions at the same time. It lowers the variance of the residuals, which narrows the bands, but it also makes the

coefficient estimates noisier, and noisier coefficients widen them. Which of the two channels reaches the reported intervals depends on how those intervals are built. Here the bands are bootstrapped by re-estimating the entire VAR on each artificial sample, so all 1210 coefficients are estimated again, draw after draw, and the uncertainty they carry is propagated into the interval rather than being conditioned away. If the estimator were unstable at $p = 12$, the re-estimated responses would vary wildly across draws and the bands would widen accordingly. The overfitting story is therefore not complete as it stands. To explain what Figure 5 shows, it has to argue that the variance channel beats the parameter uncertainty channel, which is formally the same claim De Graeve and Westermark (2025) make, only read with the opposite sign. The burden is symmetric, and it cannot be discharged by pointing at the number of parameters alone.

There is a second, more direct way of looking at the same question. If the extra lags were merely fitting noise, the first thing to disappear into the fitted part of the model would be the large and distinctive movements in the data, because those are exactly the movements a flexible model can chase. Figure 6 plots the in sample residuals of the Federal funds rate equation at $p = 4$ and at $p = 12$.

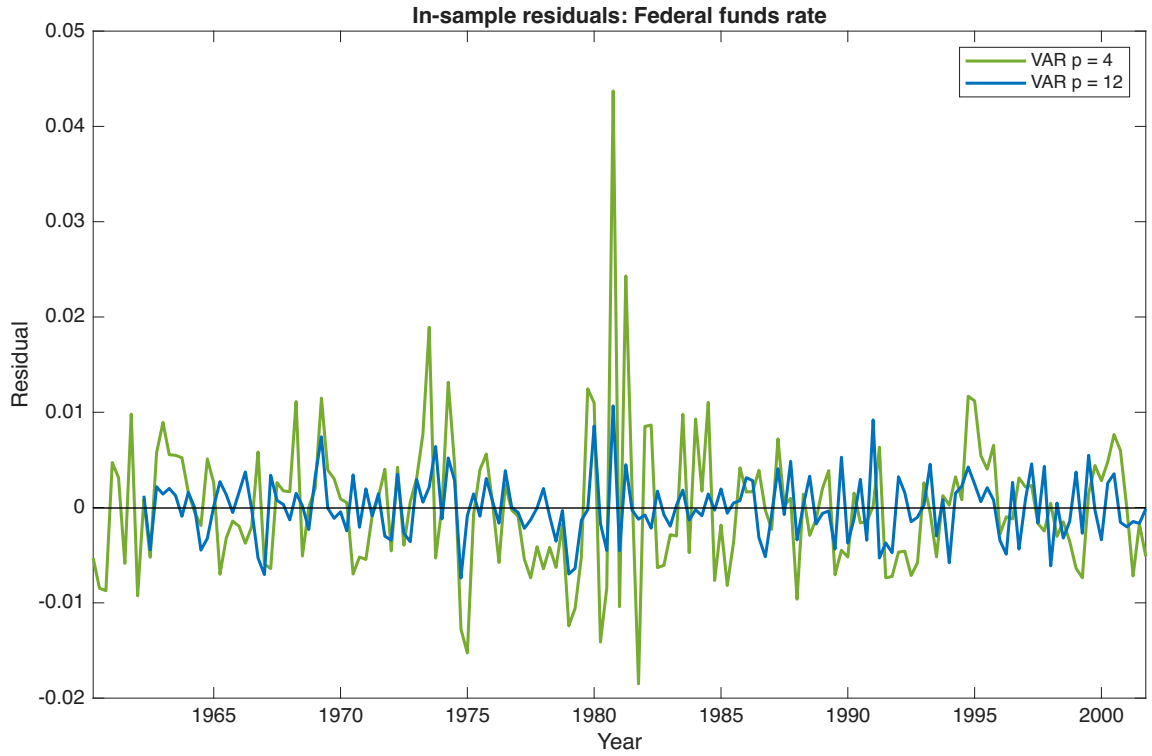


Figure 6: In-sample residuals of the Federal funds rate equation, VAR(4) against VAR(12)

The residuals of the longer VAR are visibly smaller across the whole sample, which is the variance reduction discussed above seen from the time series side rather than from the confidence bands. What survives the reduction is more interesting than its size. The single largest residual of the sample is the last quarter of 1980, in both specifications, and

the years of the Volcker disinflation remain the most turbulent stretch of the series at $p = 12$ just as they are at $p = 4$. The amplitude is much reduced, but the episode is still there and it is still the most prominent feature of the series. This is not what a model that had absorbed the data into its own parametrisation would look like. A VAR with 121 regressors per equation that were chasing noise would have had no difficulty in explaining away the most conspicuous monetary policy episode of the post war sample, and it would have done so first. The interpretation that fits the picture better is the one De Graeve and Westermark (2025) propose, that the additional lags are absorbing serial correlation that the short specification leaves in the residuals, while the genuine disturbances remain where the historical record puts them.

What the information criteria say

Altig et al. (2005) report that the AIC, HQ and SW criteria supported $q = 2$, 1 and 1 lags respectively, yet the authors set $q = 4$. Repeating the same three criteria at $p = 4, 8, 12$ in this exercise gives $p = 12$ for AIC and $p = 4$ for both HQ and SW. As already argued in Section 3, these criteria are built to minimise out-of-sample forecast error, not to guard against truncation bias, so neither Altig et al. (2005) own departure from their criteria nor the disagreement found here settles which lag order is appropriate for structural inference; it only confirms that forecasting criteria are the not rigth tool for this structural question.

5 Structural Implications

The structural estimates in this section show that the resolution of the micro–macro price conflict proposed by Altig et al. (2011) is not robust to the lag structure of the VAR it is estimated from. The mechanism fails through the very parameter it rests on: as the lag length increases the cost of varying capital utilisation collapses towards zero, firm specificity stops being an economically binding constraint, and the two versions of the model end up saying the same thing about price stickiness.

5.1 DSGE structural parameters

In Table 2 I replicate the results of Altig et al. (2005) Table 2[3] across different lag dynamics. The baseline column reproduces their estimates. The price markup λ_f is calibrated, so it carries no standard error.

[3] I report Altig et al. (2005) Table 2 in Appendix D.

Table 2: DSGE Structural Parameters Comparison Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Price markup (λ_f)	1.0100	1.0100	1.0100
	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.7221***	0.5617***	0.4838***
	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0402	0.0402	0.0129***
	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.9938	0.1175	0.0042
	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.7066***	0.6935***	0.8596***
	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.2807*	3.0372*	6.8431***
	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8077***	0.7531***	0.8346***
	(0.2111)	(0.1473)	(0.1944)

Two parameters carry the result, and I take them first. The Phillips curve slope γ is unchanged at 0.0402 between $p = 4$ and $p = 8$, and falls to 0.0129 at $p = 12$. What happens to its precision matters as much as the fall itself. The standard error goes from 0.0246 at the baseline, where the slope cannot be distinguished from zero, to 0.0042 at $p = 12$, where it is significant at the one percent level. The long-lag VAR therefore delivers an estimate of the Phillips curve slope roughly six times more precise than the baseline one, which is the variance reduction mechanism of De Graeve and Westermark (2025) appearing inside the structural estimates rather than in the reduced form.

The capacity utilisation cost σ_a moves from 1.9938 at $p = 4$ to 0.1175 at $p = 8$ and to 0.0042 at $p = 12$. This is the movement that drives everything that follows. As discussed in Section 2.2, a σ_a close to zero means that utilisation is almost free to vary, so a firm tied to its own capital can still draw whatever quantity of capital services it needs at negligible cost, and firm specificity stops being an economically binding constraint.

There is an important qualification to make about σ_a, and it strengthens rather than weakens the argument. At no lag order is σ_a statistically distinguishable from zero. Strictly, therefore, one cannot reject that σ_a was zero in ACEL's own specification. The resolution of the micro–macro conflict rests entirely on the point estimate of a parameter that the data do not pin down, and what the long-lag VAR does is to move that point estimate into precisely the region where Altig et al. (2011) themselves concede that the mechanism fails. The difference between the baseline and $p = 12$ is

not that the parameter becomes insignificant, since it never was significant, but that at $p = 12$ it is tightly bounded around zero, with a standard error of 0.0230, whereas at the baseline the same parameter was consistent with almost any value in a wide interval.

The remaining parameters move less, but one of them connects the structural estimates to the impulse responses of Section 4.1. Wage stickiness ξ_w falls from 0.7221 at the baseline to 0.4838 at $p = 12$, losing roughly a third of its baseline value, so labour market frictions become less stringent exactly where the VAR(12) delivers the hump shaped real wage response that is absent in the shorter lag versions, Figure 2. Habit formation b is stable at around 0.70 for $p = 4$ and $p = 8$ and rises to 0.86 at $p = 12$. The money demand elasticity ϵ fluctuates mildly and stays close to its baseline value of 0.81. The investment adjustment cost S'' roughly doubles, from 3.2807 to 6.8431, and becomes precisely estimated only at $p = 12$, which translates into a higher cost of adjusting the capital stock over time. It enters the non-linear expression for χ alongside σ_a, but as discussed in Section 2.2 it is not the key driver of the real rigidity.

5.2 The real rigidity parameter

Both χ and the Calvo price stickiness parameters ξ_p are implied parameters. They are not estimated directly, but recovered by inverting the Phillips curve slope γ equation (2) at the estimated parameters, so the IRF matching estimation does not attach standard errors to them. To attach a measure of uncertainty I run a Monte Carlo simulation with one million draws from the joint distribution of the estimated structural parameters. Joint normality is the assumption already used in the estimation itself, since Altig et al. (2005) recover the sampling variance of the structural parameters through the delta method, which assumes exactly that. Appendix G describes the delta method and Appendix H reports the simulated densities. The intervals reported here and in the next section are percentiles of the simulated distribution.

Table 3: Implied Strategic Complementarity Parameter χ Across Lag Lengths

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model ($\chi \equiv 1$)			
χ_{hom}	1.0000	1.0000	1.0000
	(-)	(-)	(-)
Firm-Specific Capital Model			
χ_{fs}	0.0307	0.2030	0.8671***
	(0.0596)	(0.1667)	(0.1698)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. χ is recovered by inverting $\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi$ at the estimated γ and implied ξ_p; $\chi_{hom} \equiv 1$ by construction (Proposition 1(i) in Altig et al., 2011) and carries no standard error. Standard errors for χ_{fs} are computed from the same 1,000,000-draw joint Monte Carlo used for the Calvo parameters.

Table 3 reports how the real rigidity parameter χ behaves across lag orders in the firm-specific capital model, against the homogeneous model where it is set to one by construction. At the baseline $\chi_{fs} = 0.0307$, with a 95 percent simulated interval of $[0.004, 0.146]$. The attenuation is close to total: a firm that is able to reset its price moves it by about three percent of what it would move it in the absence of the real rigidity. At $p = 8$ the parameter rises to 0.2030, and at $p = 12$ to 0.8671, with a 95 percent interval of $[0.342, 0.959]$. The baseline and the long-lag intervals do not overlap, so the movement is not an artefact of the imprecision of either estimate.

An attenuation factor of 0.03 means that firm specificity does almost all of the work in dampening price adjustment, while a factor of 0.87 means that it does almost none of it and the firm-specific model behaves nearly as the homogeneous one does. The question that matters for Altig et al. (2011) is whether χ is small enough to rationalise a low ξ_p, and at $p = 12$ it is not.

5.3 Calvo price stickiness parameters and the collapse of the firm-specific mechanism

Having in mind equation (2), the results on the Calvo price stickiness parameters reported in Table 4 follow directly from the two movements just described.

Table 4: Implied Calvo Price Stickiness Parameters Across Lag Lengths

Parameter	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous capital ($\xi_{p,hom}$)	0.8213***	0.8213***	0.8958***
	(0.0480)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3372**	0.6448***	0.8884***
	(0.1505)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors are obtained by Monte Carlo simulation with 1,000,000 draws from the joint distribution of the estimated structural parameters.

In the homogeneous capital model $\chi = 1$, so the whole burden of delivering a small γ falls on price stickiness alone. The parameter is 0.8213 at $p = 4$ and at $p = 8$, with a 95 percent simulated interval of $[0.747, 0.936]$ at the baseline, and it rises to 0.8958 at $p = 12$, with an interval of $[0.869, 0.923]$. The flatter Phillips curve of the long-lag VAR therefore makes the homogeneous model worse on its own terms. The friction it requires in order to match the aggregate evidence becomes stronger, not weaker, and it was already implausibly strong at the baseline.

In the firm-specific model the parameter moves a great deal more. It is 0.3372 at the baseline, with a 95 percent interval of $[0.057, 0.709]$, rises to 0.6448 at $p = 8$, with an interval of $[0.413, 0.853]$, and reaches 0.8884 at $p = 12$, with an interval of $[0.815, 0.898]$. The baseline point estimate lies outside the 95 percent interval of the $p = 8$ distribution, where only 1.1 percent of the draws fall at or below it. At $p = 12$ the separation is complete, since no draw at all falls at or below the baseline value. Whatever the firm-specific model says about price stickiness at $p = 4$, it is not what the same model says once the VAR is given a richer lag structure.

5.4 Implied duration of prices

Under Calvo price setting the implied duration, average time between re-optimisations, is $D = \frac{1}{1-\xi_p}$, and it is here that the size of the gap between the two models becomes meaningful.

Table 5: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

Table 5 shows how increasing the lag length of the VAR used to estimate the Altig et al. (2011) DSGE brings back the micro–macro price duration conflict that firm-specific capital was introduced to resolve. At the baseline the firm-specific model implies that prices are re-optimised on average every 1.5 quarters, against 5.6 quarters in the homogeneous model, and it is this difference that allows ACEL to claim that only the firm-specific version is compatible with the micro evidence. At $p = 8$ the firm-specific duration almost doubles, to 2.8 quarters, which is still half the homogeneous figure but already a long way from the micro evidence. At $p = 12$ the two models imply 9.0 and 9.6 quarters.

The homogeneous capital model, whose implied duration rises from 5.6 to 9.6 quarters, was already in open conflict with the micro evidence at the baseline, and now ends up requiring prices to remain fixed for well over two years. The long-lag results therefore do not point at the homogeneous capital model at the expense of the firm-specific one. They push both models into the same implausible region, and the conclusion is not that one version wins the argument, but that the reconciliation Altig et al. (2011) offer is not available at all once the VAR is allowed a richer lag structure.

This outcome is the one their own argument anticipates. The reconciliation works only if varying capital utilisation is costly enough to make firm specificity bind, and the models' authors state that "to the extent that capital utilization rates can easily be varied, the assumption of firm-specific capital loses its ability to rationalize low values of ξ_p" (Altig et al., 2011, p. 227). The long-lag VAR drives σ_a to essentially zero, which is exactly the case they describe. I do not claim that $p = 12$ delivers the true structural parameters, and nothing here establishes that the firm-specific mechanism is wrong in itself. What the exercise establishes is that the conclusion drawn from it is not robust to a choice, the lag order of the VAR, that was not treated as a substantive modelling decision.

6 Conclusion

This work replicates Altig et al. (2005) and extends the lag length of the VAR whose impulse responses identify their DSGE, following the line of work opened by De Graeve and Westermark (2025) and Baumeister (2025).

Their main argument is that a VAR with few lags suffers from truncation bias, a form of misspecification that distorts the impulse responses at every horizon. Even where long-lag VARs are known to have good asymptotic properties (Plagborg-Møller and Wolf, 2021), they are widely treated as unworkable in finite samples, since the extra parametrisation that reduces the truncation bias is presumed to bring greater uncertainty around the estimates. De Graeve and Westermark (2025) show that this is only part of the story: correcting the misspecification also lowers the variance of the residuals, and that works against the imprecision coming from the additional parameters. Neither paper reduces long-lag VARs to a robustness exercise. What they recommend is estimating with long lags in the first place, and the demand for robustness is only the fallback left to a researcher who insists on a short specification, who then carries the burden of showing that the result survives richer lag dynamics.

It does not survive here. The capacity utilisation cost σ_a falls from 1.9938 at $p = 4$ to 0.0042 at $p = 12$, and since σ_a is the only estimated parameter that governs the real rigidity χ in a quantitatively relevant way, its collapse removes the attenuation on which firm specificity depends. The implied duration of price contracts, 1.5 quarters under firm-specific capital against 5.6 under homogeneous capital at the baseline, becomes 9.0 and 9.6 quarters at $p = 12$. The reconciliation with the micro evidence is gone, and the confidence bands narrow rather than widen as it disappears.

This is a preliminary and exploratory exercise, and it is confined to a frequentist medium-scale VAR used as a source of moments to estimate a DSGE. A natural next step is to ask how long-lag VARs behave in a comparable setting once the parametrisation is disciplined differently, through shrinkage techniques such as ridge or lasso, or through priors in a Bayesian setting.

Despite the limitation of this work, this study draws attention to the choice of lag length, which in applied macroeconomics is usually settled by forecasting information criteria or by convention. That choice is not innocuous. Here it decides whether the micro–macro reconciliation of Altig et al. (2011) stands, much as in Baumeister (2025) it decides whether the long-run conclusions of Jordà et al. (2024) stand at all.

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A Data

Nothing in this appendix has been changed from the original source. The description of the data below reproduces, in my own words, what Altig et al. (2005, fn. 15) report in their footnote 15.

The ten series in Y_t, equation (6), are the exact same data as Altig et al. (2005). With the exception of the price of investment and of the transactions balances measure, all series are taken from the DRI Basic Economics Database. Table 6 lists the underlying series and the transformation applied to each to obtain the corresponding element of Y_t.

Table 6: Data sources and transformations

Element of Y_t	**Source series**	**Transformation**
Relative price of investment	Total investment price series of Fisher (2006)	Log difference
GDP per hour (labour productivity)	GDPQ (real, chain-weighted output); LBMNU (non-farm business hours, per capita)	Log difference
GDP deflator	GDP / GDPQ (ratio of nominal to real output)	Log difference
Capacity utilisation	IPXMCA (manufacturing capacity index)	Level
Hours	LBMNU, converted to per capita terms using P16	Log level
GDP per hour less the real wage	GDPQ, LBMNU; LBCPU (nominal hourly compensation) deflated by GDP/GDPQ	Log difference
Consumption share of GDP	GCN (nondurables) + GCS (services) + GGE (government expenditure), over GDP	Log ratio
Investment share of GDP	GCD (consumer durables) + GPI (gross private domestic investment), over GDP	Log ratio
Federal funds rate	FYFF	Level, annualised percentage points
Log velocity of MZM	GDP, GDP/GDPQ, MZM (see below)	Log level

Note: All mnemonics refer to the DRI Basic Economics Database, with the exception of the price of investment and of MZM. Output, investment, consumption and money are converted to per capita terms using P16, the DRI series for the US population aged sixteen and over, before entering the ratios and log differences above.

Two series depart from the Database. The price of investment is the total investment price series constructed by Fisher (2006), built directly from the relative price of new equipment and structures rather than recovered as an implicit deflator from nominal and real output. The transactions balances measure, MZM, is obtained by splicing two Federal Reserve Bank of St. Louis series, the M2 minus small time deposits series before 1974 and the MZM series from 1974 onward, with no rescaling at the splice point since the two series are essentially identical in that year. Altig et al. (2005) adopt MZM over narrower

aggregates such as M1 or the monetary base for two reasons. First, MZM is constructed as a measure of transactions balances, which is the natural empirical counterpart to the model's money variable Q_t. Second, the estimation strategy requires the velocity of money to be stationary, an assumption that is reasonable for MZM but considerably more doubtful for narrower aggregates.

The sample is quarterly, 1959:Q2 to 2001:Q4, giving 171 observations before any lags are consumed. At the baseline lag order $p = 4$ the first four quarters are used as initial conditions, leaving 167 usable observations; at the longer lag orders considered in this thesis correspondingly more quarters are consumed in the same way, as reported in Table 1.

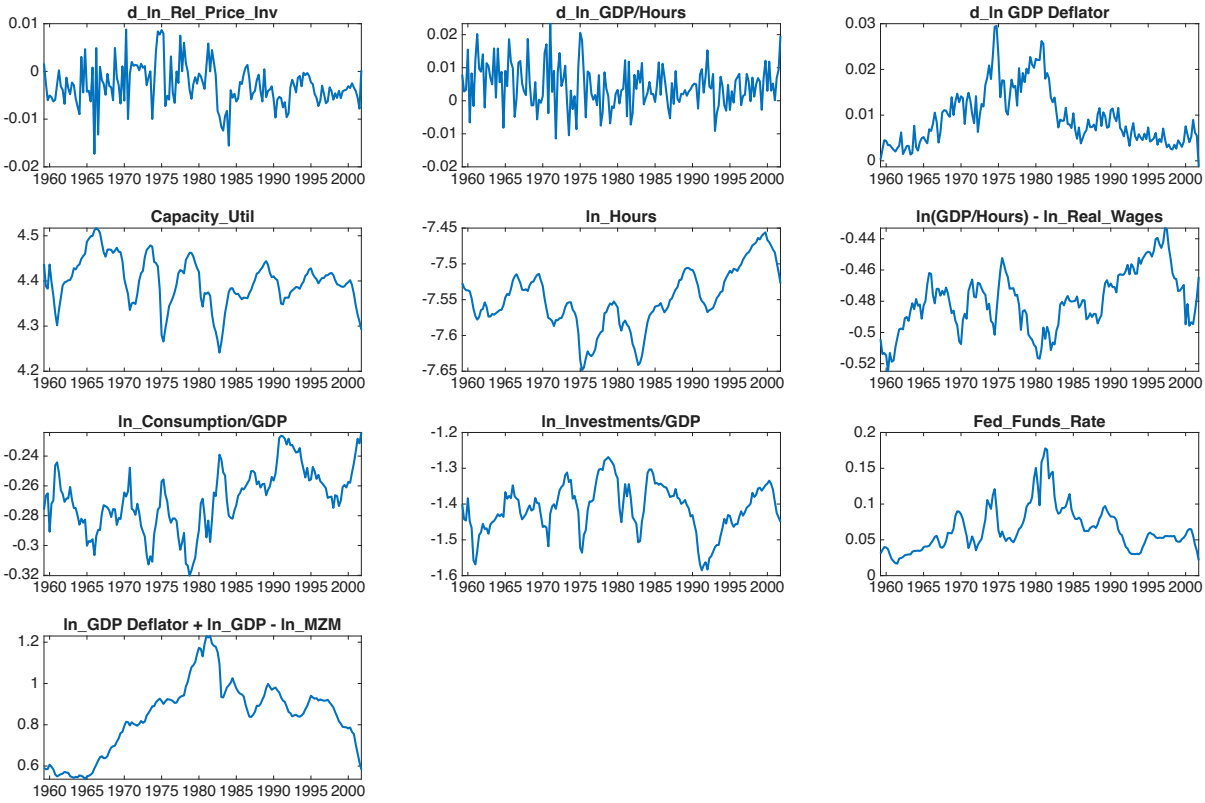


Figure 7: Raw historical series

B Identification strategy

Equation (5) also shows why identification requires restrictions. Σ_u has $K(K+1)/2$ distinct elements, but B_0^{-1} has K^2 free entries, so $K(K-1)/2$ restrictions on B_0 are needed before B_0^{-1} can be recovered from Σ_u. Being numerically sufficient is not enough: the restrictions have to be economically motivated, and it is exactly this requirement that the DSGE is used to satisfy.

C The ACEL Model: Full Derivation

This appendix reports the two model economies of Altig et al. (2011), the homogeneous and the firm-specific capital versions. Section 2.2 already gives the economic content of the comparison between the two, and of the parameters that matter for it, σ_a, S'' and χ; what follows here is a description of the equilibrium conditions from which that comparison is derived, kept to the equations needed to see where σ_a, S'' and χ enter and where the two models coincide or part company. The exposition follows Altig et al. (2011) section 2, and I keep their notation throughout.

C.1 The homogeneous capital model

The economy is populated by goods-producing firms, households and the government.

Final and intermediate good firms

A perfectly competitive final good firm combines a continuum of intermediate goods $i \in [0,1]$ using the CES technology

$$Y_t = \left[\int_0^1 y_t(i)^{\frac{1}{\lambda_f}} \, di \right]^{\lambda_f}, \qquad 1 \leq \lambda_f < \infty, \tag{14}$$

which gives the usual downward-sloping demand curve $y_t(i)/Y_t = (P_t(i)/P_t)^{-\lambda_f/(\lambda_f - 1)}$ faced by each intermediate good producer, and a price index for P_t of the same CES form. Intermediate good i is produced by a monopolist,

$$y_t(i) = K_t(i)^{\alpha} (z_t h_t(i))^{1-\alpha} - \phi z_t^*, \qquad z_t^* = \Upsilon_t^{\frac{\alpha}{1-\alpha}} z_t, \tag{15}$$

where $h_t(i)$ and $K_t(i)$ are labour and capital services, z_t is a neutral technology shock and Υ_t a capital-embodied technology shock, both following exogenous stationary processes; α is the capital share and ϕ a fixed cost calibrated so that profits are zero along the balanced growth path. Firms rent capital and labour in competitive factor markets and borrow the wage bill in advance. Prices are set on a Calvo scheme: each period a firm can re-optimise its price only with probability $1 - \xi_p$, and a firm that cannot re-optimise indexes to lagged inflation, $P_t(i) = \pi_{t-1} P_{t-1}(i)$. A firm that can re-optimise chooses $P_t(i)$ to maximise the expected discounted value of net cash flow, revenue less the wage bill, inclusive of the interest cost of borrowing it, less the cost of capital services, subject to the demand curve and to the Calvo friction on future repricing dates.

Households

Households have habit-persistent preferences over consumption and disutility of hours,

$$\mathbb{E}_t^j \sum_{l=0}^{\infty} \beta^l \left[\log(C_{t+l} - bC_{t+l-1}) - \psi_L \frac{h_{j,t+l}^2}{2} \right], \tag{16}$$

and choose consumption, investment, money and bond holdings subject to a budget constraint that combines labour income, capital rental income, firm profits and a cash-in-advance style transactions cost on consumption purchases, $[1 + \eta(V_t)]P_tC_t$, where velocity is $V_t = P_tC_t/Q_t$. Two cost functions inside that budget constraint carry the two structural parameters this thesis focuses on. First, capital services are $K_t = u_t\bar{K}_t$, where $\bar{K}_t$ is the physical stock and u_t the utilisation rate, and raising utilisation is costly in units of the final good, $a(u_t)\bar{K}_t$, with a increasing and convex; the curvature of a in steady state,

$$\sigma_a = \frac{a''(1)}{a'(1)} \geq 0, \tag{17}$$

is estimated, normalising $u_t = 1$ and $a(1) = 0$ in steady state. Second, physical capital accumulates subject to an adjustment cost on the flow of investment,

$$\bar{K}_{t+1} = (1 - \delta)\bar{K}_t + \left[1 - S\left(\frac{I_t}{I_{t-1}} \right) \right] I_t, \tag{18}$$

with S increasing, convex and $S = S' = 0$ in steady state; its curvature,

$$S'' > 0, \tag{19}$$

is also estimated. Neither σ_a nor S'' affects the non-stochastic steady state, only the dynamics, which is why both can only be pinned down from impulse responses rather than from steady-state ratios.

Each household is also a monopoly supplier of a differentiated labour service, aggregated into economy-wide labour H_t through the same CES structure as the goods market, and wages are set under an analogous Calvo friction, with re-optimisation probability $1 - \xi_w$ and indexation to $\pi_{t-1}\mu_{z^*}$ for households that cannot reset. Monetary policy is specified as a decomposition of money growth into a systematic response to the two technology shocks and an exogenous, autoregressive monetary policy shock $\varepsilon_{M,t}$, and Christiano et al. (2005) show that impulse responses under this rule closely match those from an explicit Taylor rule. The government runs a balanced intertemporal budget through lump-sum taxes. Equilibrium closes with loan market clearing, $W_tH_t = x_tM_t - Q_t$, and the aggregate resource constraint, $[1 + \eta(V_t)]C_t + \Upsilon_t^{-1}I_t + a(u_t)\bar{K}_t \leq Y_t$, and is solved by log-linear approximation around the non-stochastic steady state.

C.2 The firm-specific capital model

Every equation above is unchanged except that firms now own their capital and cannot adjust their stock within the period: firm i's capital accumulates according to its own version of (18), and it can only vary the capital services it uses by changing its own utilisation rate $u_t(i)$, at the cost $a(u_t(i))\bar{K}_t(i)$. There is no market on which the firm can rent additional capital services from anyone else, so σ_a now governs not an economy-wide margin but each firm's own, private cost of meeting extra demand. The firm chooses prices, employment, investment and utilisation to maximise the discounted value of net cash flow, subject to the same timing and Calvo pricing friction as in the homogeneous model.

C.3 Implications for inflation

The equilibrium conditions of the two models coincide except for the equation governing aggregate inflation, which takes the form

$$\Delta\hat{\pi}_t = \mathbb{E}\big[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t \mid \Omega_t\big], \qquad \gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\,\chi, \tag{20}$$

reproducing equations (1) and (2) of Section 2.2, where Δ is the first-difference operator, $\hat{s}_t$ economy-wide real marginal cost, and Ω_t the information set containing the technology but not the monetary policy shock. Altig et al. (2011) establish that $\chi = 1$ under homogeneous capital, and that χ is a non-linear function of the model's parameters, principally σ_a and S'', under firm-specific capital. Both models are parametrised in terms of γ rather than ξ_p, so they carry an identical list of parameters and, for given parameter values, are observationally equivalent in aggregate prices and quantities. This is what lets the estimation in this thesis, as in Altig et al. (2011), stay agnostic about which version of the model generated the data: the two are told apart only afterwards, from the value of ξ_p that a given estimated γ implies once χ is fixed at one or computed from the non-linear mapping, which is the exercise carried out in Sections 2.2, 5.1 and 5.4.

D ACEL 2005 WP Table 2

I report Table D, which is the exact reproduction of the Table 2 reported in Altig et al. (2005) at page 41.

Table 7: Estimated parameter values, ζ_1

Model	λ_f	ξ_w	γ	σ_a	b	S''	ϵ
Benchmark	1.01	0.72 (0.13)	0.040 (0.020)	2.02 (2.32)	0.70 (0.04)	3.28 (1.69)	0.80 (0.21)
Monetary Shocks Only	1.08 (0.10)	0.62 (0.07)	0.106 (0.061)	0.01 (0.01)	0.73 (0.05)	3.63 (1.73)	0.82 (0.19)
Neutral Technology Shocks Only	2.05 *(n.a.)*	0.93 *(n.a.)*	9.5e+006 *(n.a.)*	2.3e+046 *(n.a.)*	0.054 *(n.a.)*	1.00 *(n.a.)*	0.97 *(n.a.)*
Embodied Technology Shocks Only	1.01	0.70 (0.36)	0.033 (0.020)	5.84 (8.03)	0.23 (0.29)	0.90 (0.73)	1.24 (0.71)
Low Cost of Varying Capital Utilization	1.46 (1.93)	0.69 (0.12)	0.050 (0.031)	.01	0.69 (0.05)	2.79 (1.47)	0.79 (0.21)
Intermediate Markup	1.04	0.72 (0.13)	0.042 (0.022)	1.91 (2.23)	0.70 (0.04)	3.18 (1.65)	0.80 (0.21)
High Markup	1.20	0.69 (0.12)	0.050 (0.031)	1.46 (1.93)	0.69 (0.05)	2.79 (1.47)	0.79 (0.21)
Benchmark (6 lag VAR)	1.01	0.93 (0.07)	0.152 (0.229)	10.54 (28.11)	0.72 (0.06)	2413.49 (7368.62)	0.81 (0.20)

Note: This table reproduces Altig et al. (2005, Table 2) exactly as printed there; it has not been re-estimated or altered in any way. Standard errors in parentheses. *(n.a.)* marks a case for which the minimum-distance step did not converge to a well-behaved covariance matrix, so no standard error is reported.

E IRFs with all confidence intervals

In Figure 8 and 9 are reported the IRFs images that include the 95% confidence interval of every lag choice, not only for the baseline VAR(4).

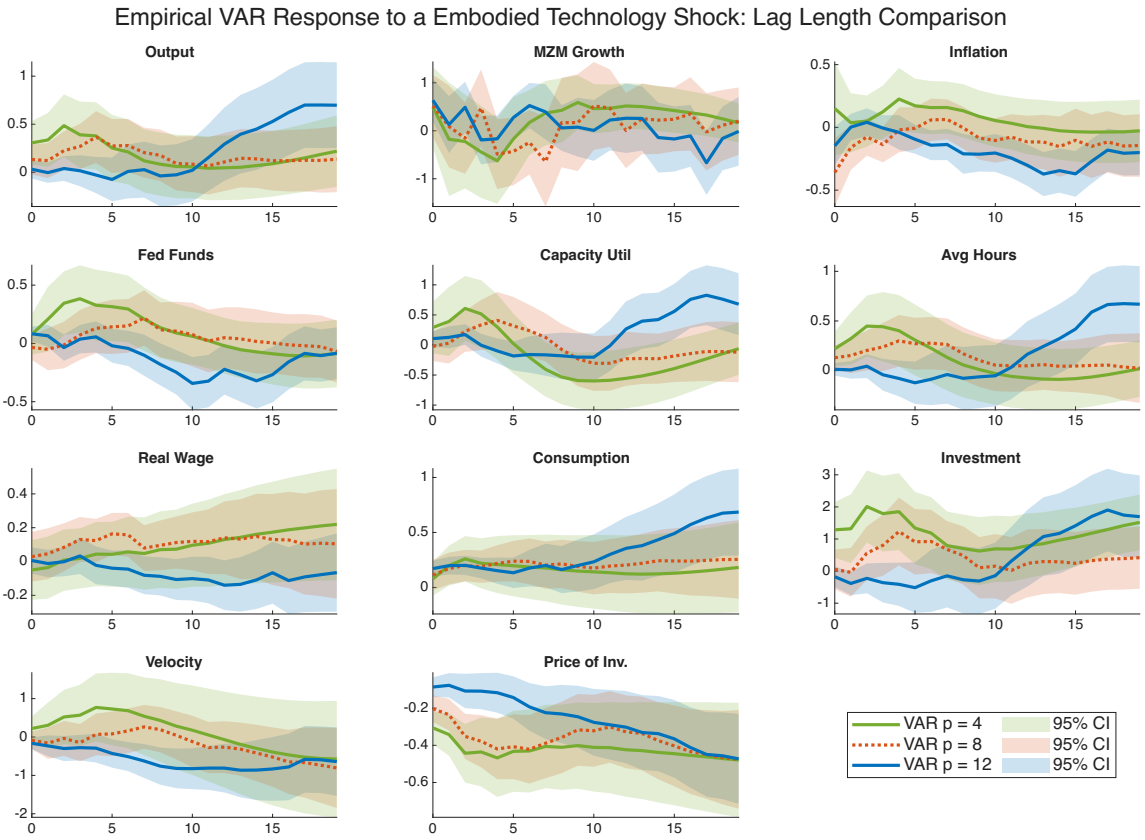


Figure 8: VARs' IRFs to a Embodied Technology shock across different lag structure, 95% CI

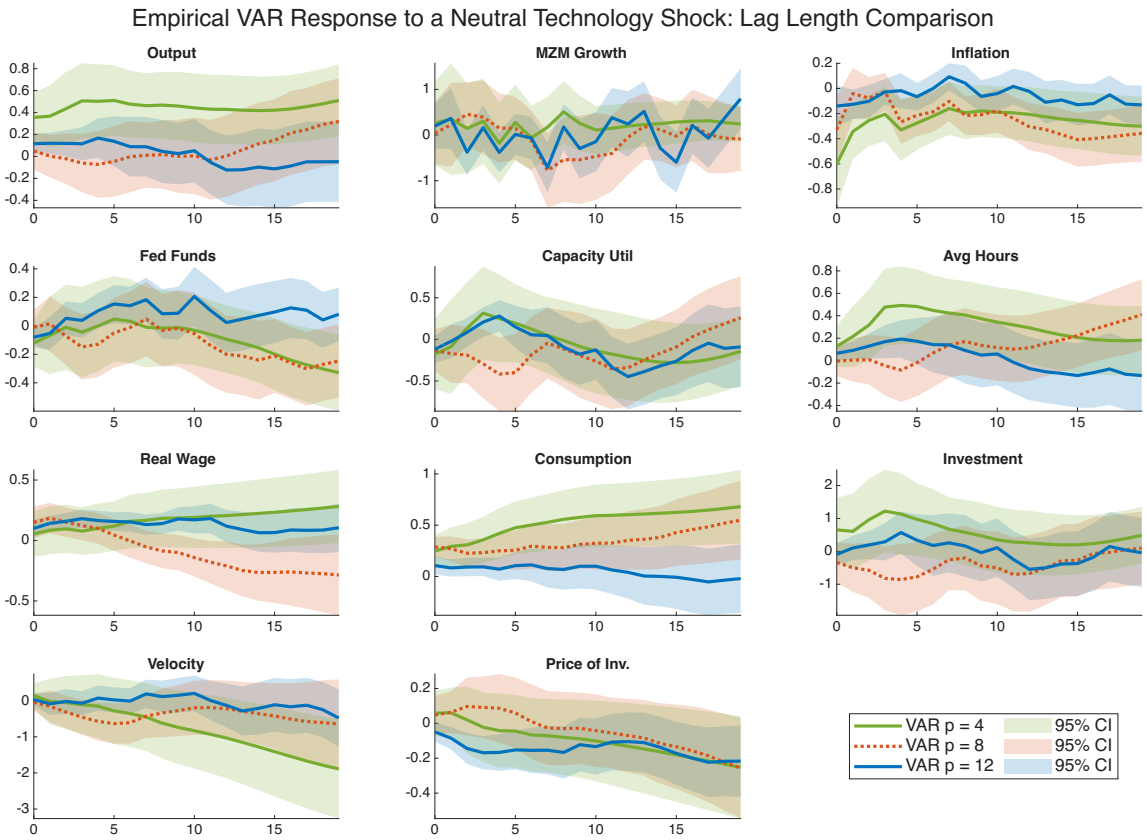


Figure 9: VARs' IRFs to a Neutral Technology shock across different lag structure, 95% CI

F Residuals

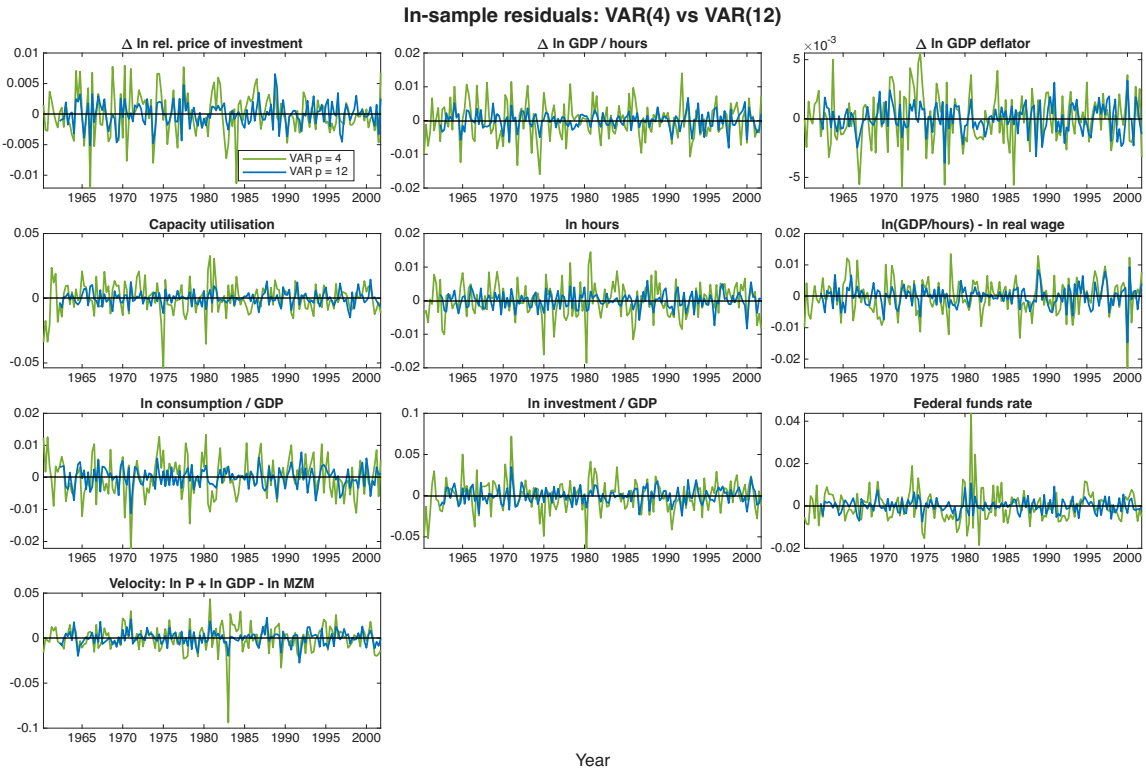


Figure 10: In-sample residuals of all the variables, VAR(4) against VAR(12)

G Delta Method

Following the standard error estimation strategy for the DSGE structural parameters chosen by Altig et al. (2005) and well explained in their footnote number 14.

Given the IRF minimum distance estimator Equation 7, let the criterion be denoted as $L(\zeta, \hat{\Psi}) \equiv (\hat{\Psi} - \Psi(\zeta))' V^{-1} (\hat{\Psi} - \Psi(\zeta))$ so that $L_1(\hat{\zeta}, \hat{\Psi}) = 0$.

Note that the mapping in 7 is given by $\hat{\zeta} = f(\hat{\Psi})$. In order to obtain the sampling variance of the estimator, $\hat{\zeta}$, as a function of the sampling variance of $\hat{\Psi}$, the delta function method approximates $f(\hat{\Psi})$ by its linear expansion about the true value of Ψ, Ψ^0. The Taylor's expansion is given by, $f(\hat{\Psi}) \approx f(\Psi^0) + f'(\Psi^0)(\hat{\Psi} - \Psi^0)$. Here $f(\Psi^0) = \zeta^0$, where ζ^0 is the true unknown value of ζ, by the consistency of the estimator.

Then $\sqrt{T}(\hat{\zeta_T} - \zeta^0)$ is asymptotically Normally distributed with zero mean and variance equal to $f'(\Psi^0) W f'(\Psi^0)^T$, where the superscript T indicates the transposition operator, and W is the asymptotic variance-covariance matrix of $\sqrt{T}(\hat{\Psi} - \Psi)$.

Then, implicit function theorem is used to approximate $f'(\Psi^0)$ by $-L_{11}(\zeta^0, \Psi^0)^{-1} L_{12}(\zeta^0, \Psi^0)$. Then, V is not selected as random, in practice a diagonal matrix is built using the VAR IRFs variances W. In addition, W is replaced by its sample estimate, as are ζ^0 and Ψ^0 in

the expression for f'.

H Monte Carlo Inference on the Calvo Parameters

Neither χ nor the Calvo price stickiness parameters ξ_p are estimated directly. They are recovered by inverting the non-linear mapping in equation (2) at the estimated structural parameters, so the IRF matching estimator attaches no standard error to either of them. The delta method of Appendix G gives the joint sampling variance-covariance V_{DSGE} of the estimated structural parameters, but propagating that uncertainty through a non-linear mapping analytically is not tractable here, so I do it by simulation.

I draw the full vector of structural parameters jointly from a multivariate normal distribution centred at the point estimates $\hat{\zeta}$, with variance-covariance V_{DSGE}, for $N = 1{,}000{,}000$ draws. Drawing the parameters jointly, rather than varying γ alone, matters because χ and ξ_p depend on several parameters at once, and those parameters are themselves correlated: the simulation has to preserve the correlation structure encoded in V_{DSGE}, not just the marginal uncertainty of each parameter taken on its own.

Five of the estimated parameters enter the mapping directly: the Phillips curve slope γ, the capacity utilisation cost σ_a, the investment adjustment cost S'', habit formation b, and the money demand elasticity ϵ. The remaining calibrated parameters are held fixed at their original values. For every draw of these five parameters I recover the implied Calvo parameters using the "findksip" routine from the Altig et al. (2005) replication code. In the homogeneous capital model $\chi \equiv 1$ by construction, so equation (2) collapses to a quadratic in $\xi_{p,hom}$ and the routine solves it in closed form. In the firm-specific model χ is itself a non-linear function of the draw, so the routine searches a grid of candidate values of $\xi_{p,fs}$, brackets the point at which the implied γ crosses the target value, and refines the root numerically.

Not every draw returns an economically meaningful solution. Some draws imply a γ for which no admissible root exists, or a complex valued one, and these are discarded rather than recorded as zero or as a boundary value. The share of draws that return a valid solution is 78.6% at $p = 4$, 31.8% at $p = 8$, and 49.2% at $p = 12$. The empirical distributions of $\xi_{p,fs}$ and $\xi_{p,hom}$ across the valid draws, plotted below, are what the percentile intervals reported in Sections 5.2 and 5.3 are computed from.

Here I plot the probability density functions (PDF) and the cumulative density functions (CDF) derived from the Monte Carlo simulation.

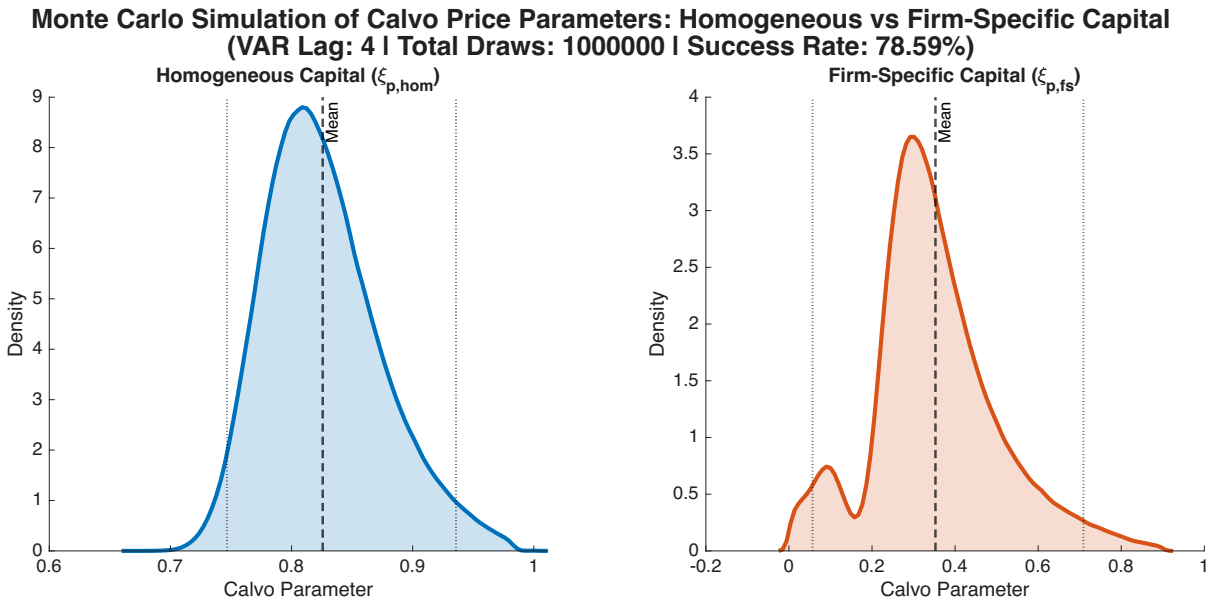


Figure 11: Monte Carlo PDF $p = 4$

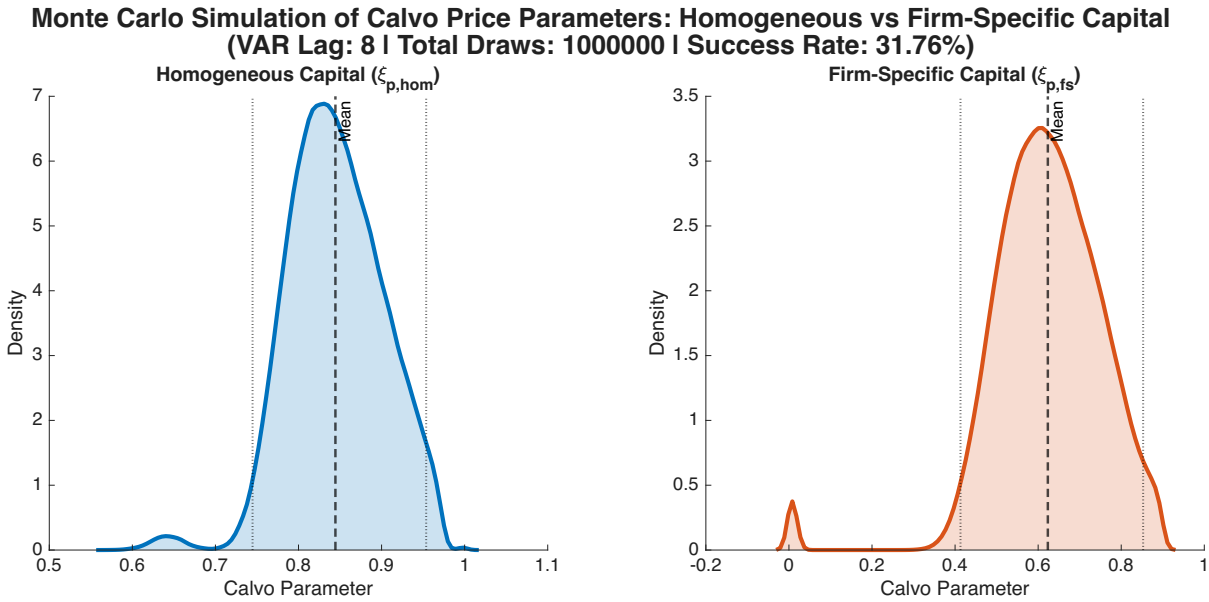


Figure 12: Monte Carlo PDF $p = 8$

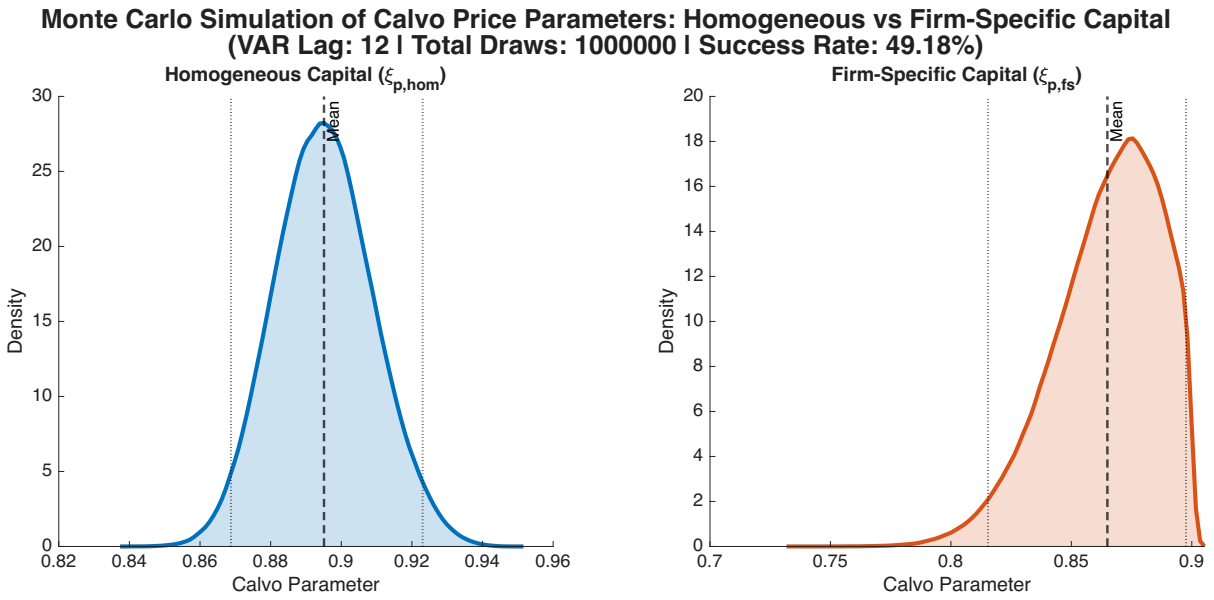


Figure 13: Monte Carlo PDF $p = 12$

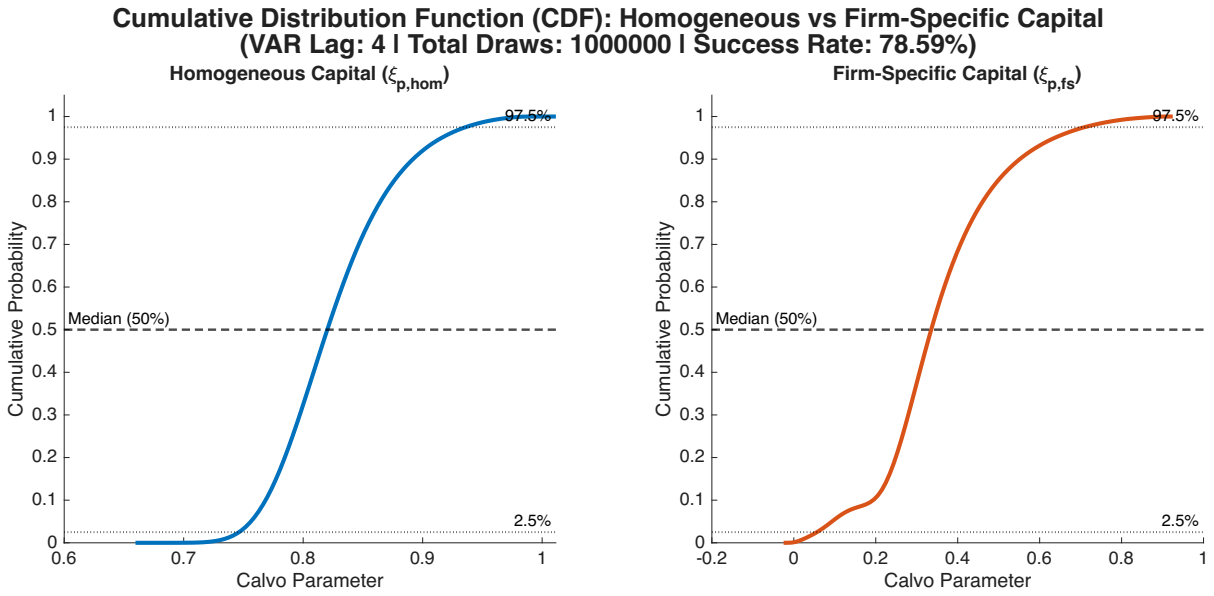


Figure 14: Monte Carlo CDF $p = 4$

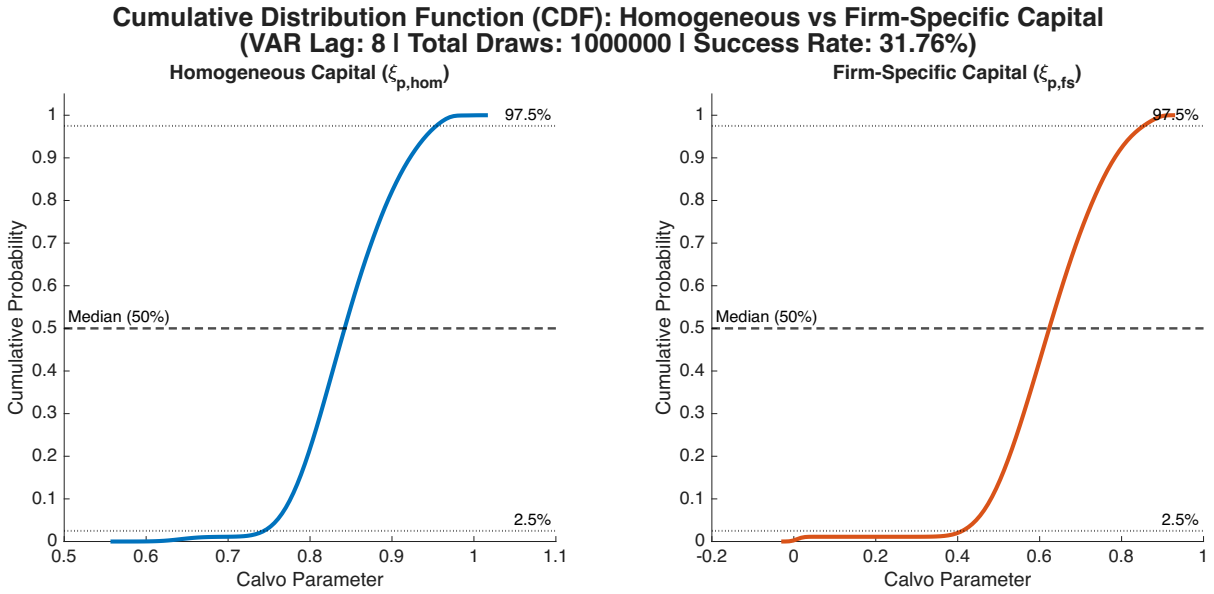


Figure 15: Monte Carlo CDF $p = 8$

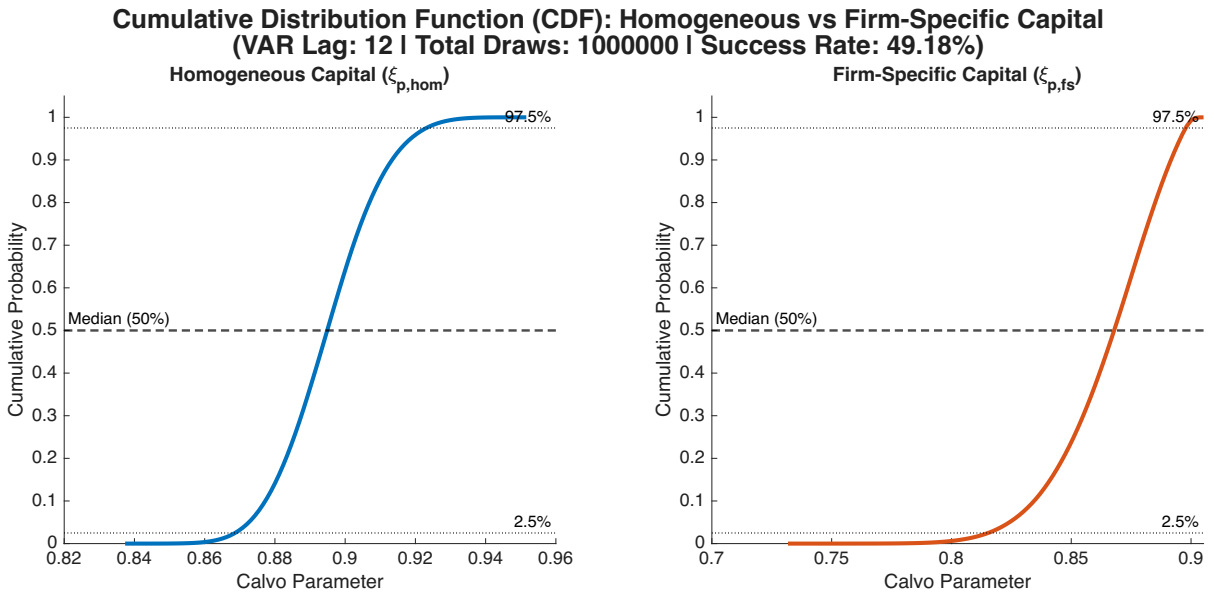


Figure 16: Monte Carlo CDF $p = 12$

I VAR Aligned

The baseline results in the main text use, for every lag order, all the observations available to that specification, so the VAR(4) is estimated on more data than the VAR(12). This appendix checks whether that choice matters, by re-estimating every VAR on the sample common to all three lag orders, the one available to the VAR(12), and comparing the resulting impulse responses and structural parameters against the full-sample baseline reported in the main text.

Impulse responses

Figures 17 and 18 compare the aligned and full-sample responses to a monetary policy shock, with their own 95 percent bootstrap confidence bands, at $p = 4$ and $p = 8$.

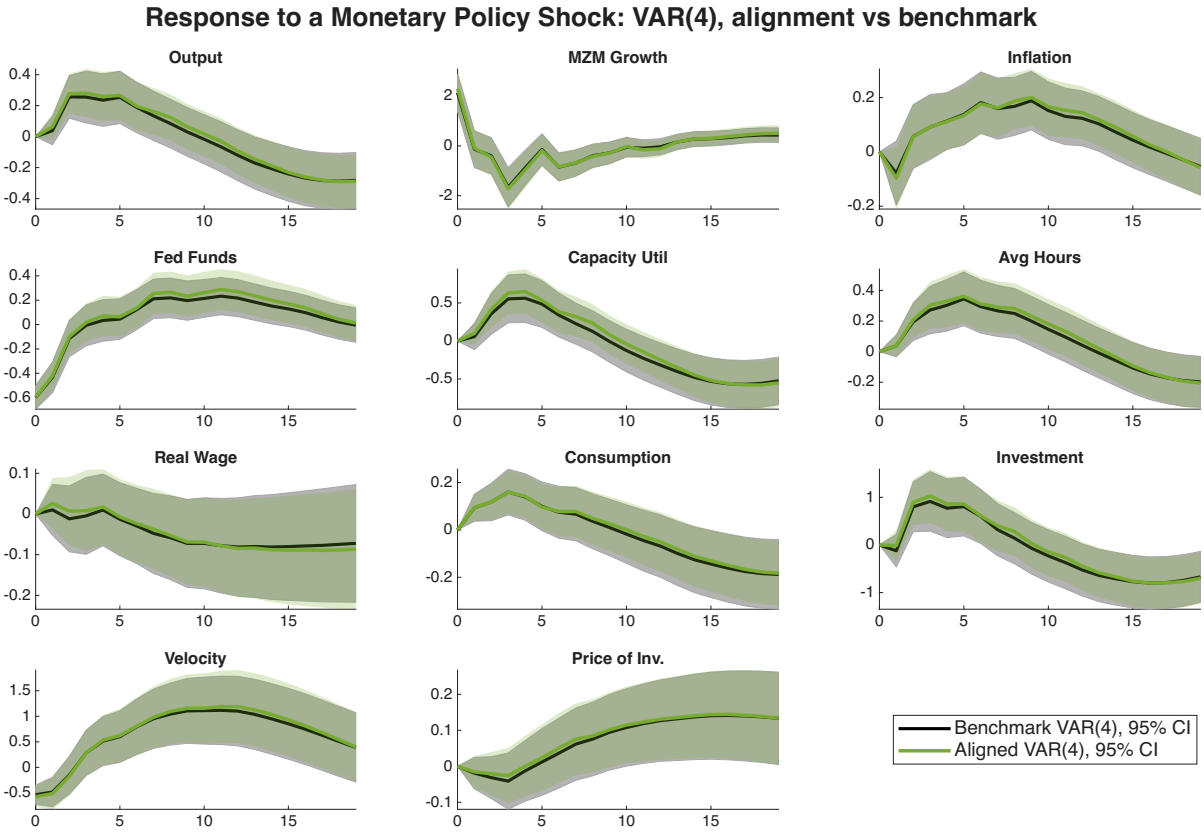


Figure 17: VAR aligned monetary policy shock IRFs comparison $p = 4$

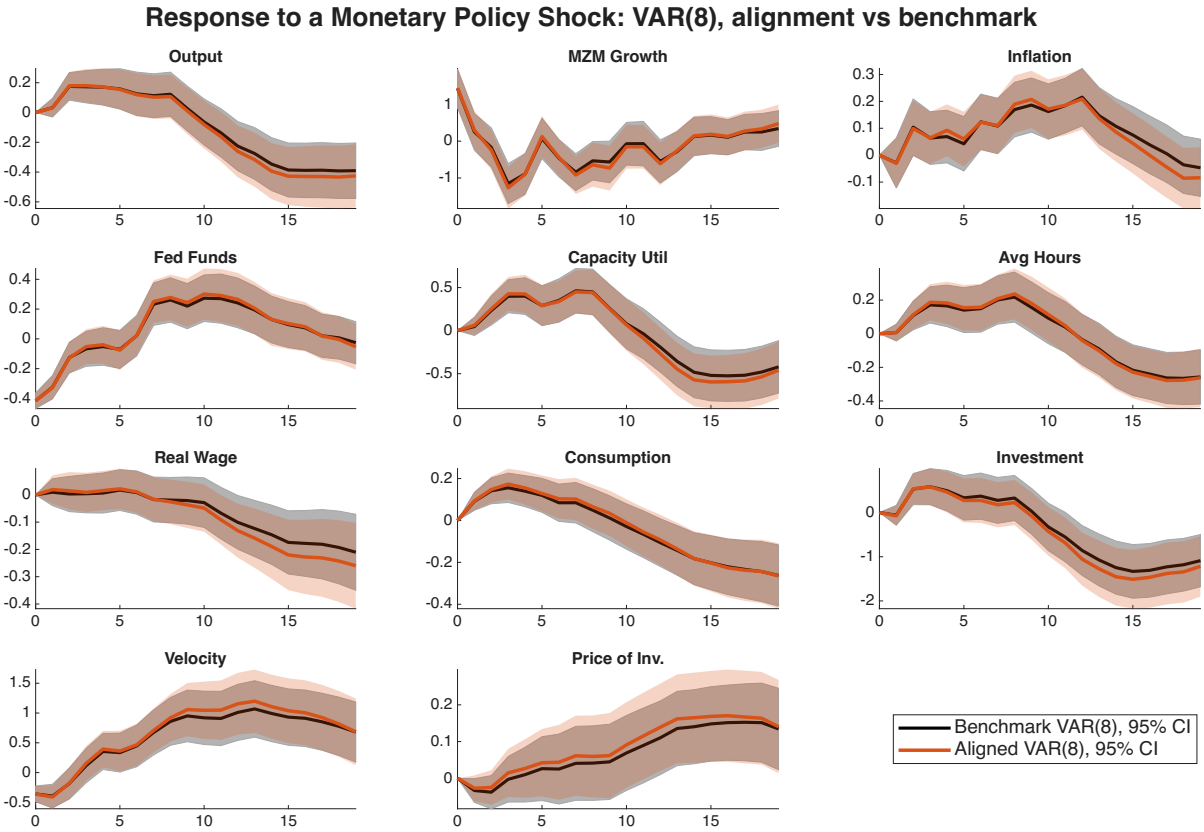


Figure 18: VAR aligned monetary policy shock IRFs comparison $p = 8$

Having fewer observations than the baseline makes no visible difference to the monetary policy shock IRFs at $p = 4$. At $p = 8$ there is some difference in the point estimates beyond horizon $h > p$, but it remains small relative to the size of the responses themselves.

Figures 19 and 20 repeat the comparison for the embodied technology shock, and Figures 21 and 22 for the neutral technology shock.

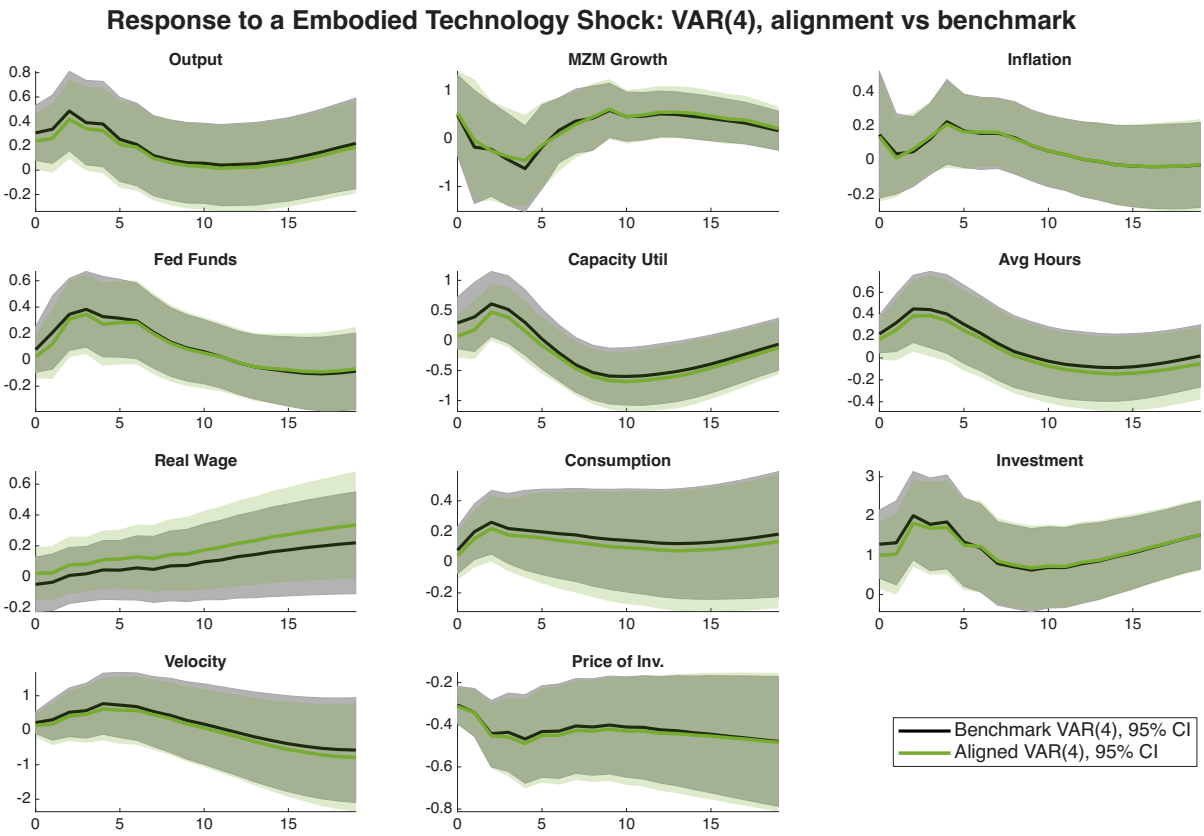


Figure 19: VAR aligned embodied technology shock IRFs comparison $p = 4$

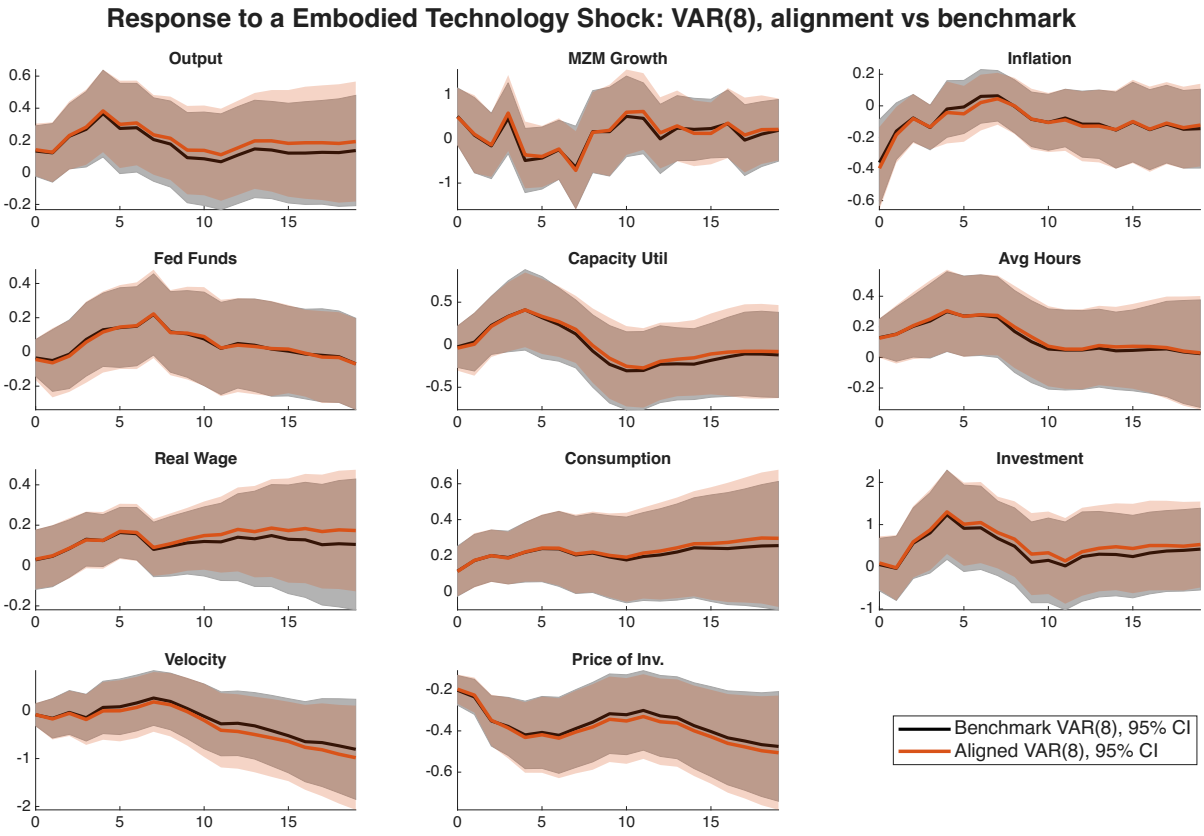


Figure 20: VAR aligned embodied technology shock IRFs comparison $p = 8$

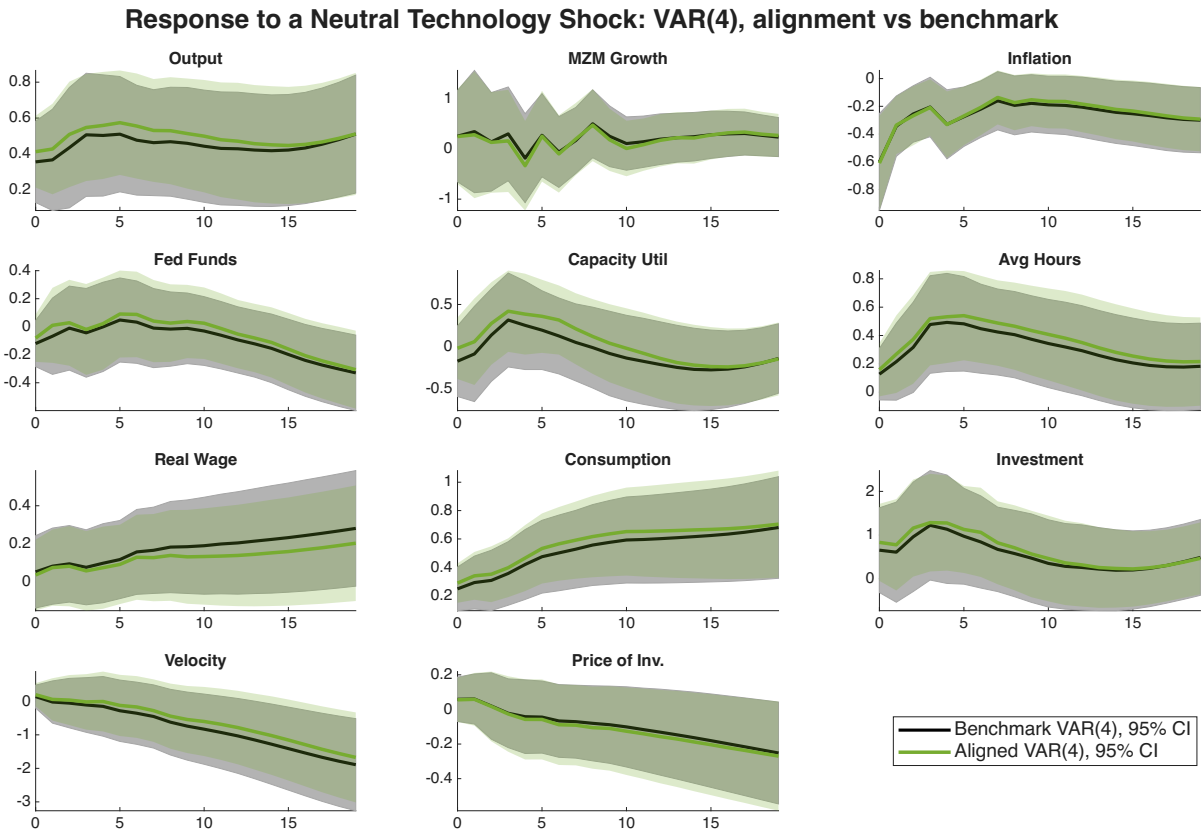


Figure 21: VAR aligned neutral technology shock IRFs comparison $p = 4$

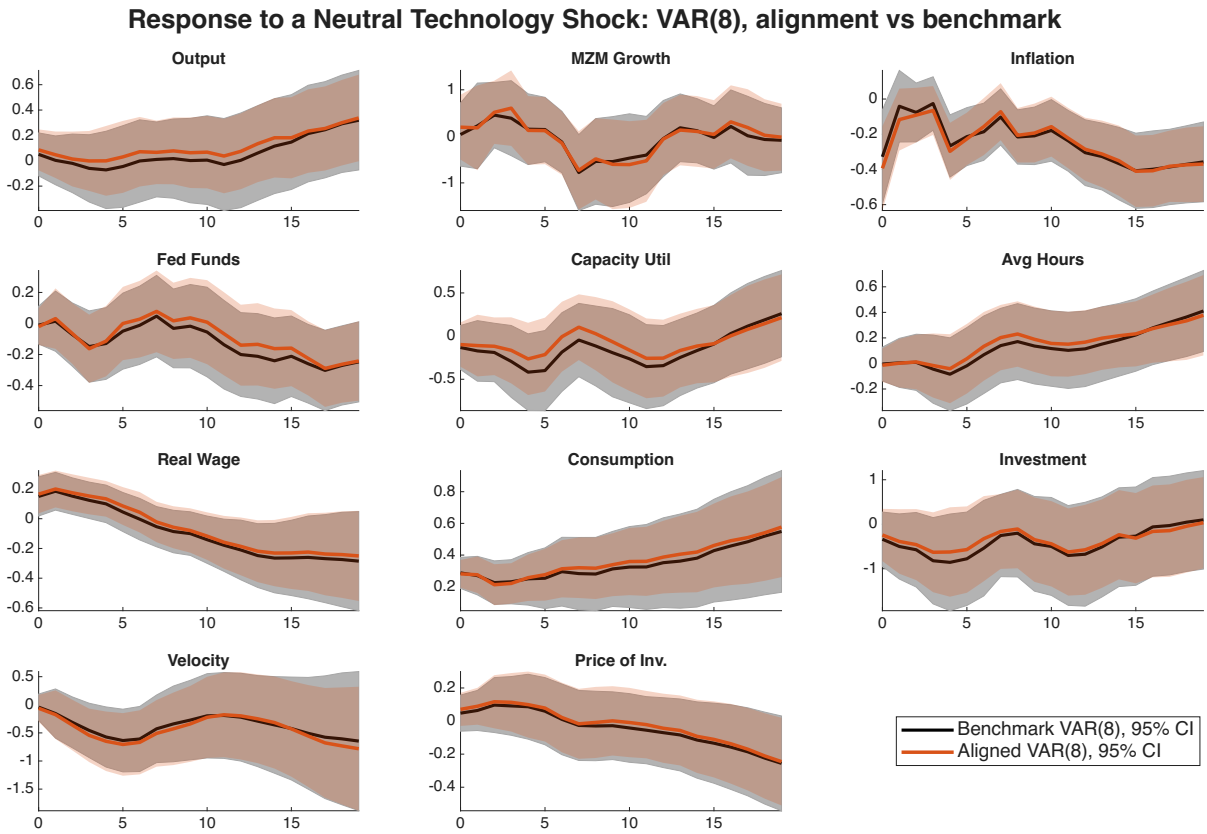


Figure 22: VAR aligned neutral technology shock IRFs comparison $p = 8$

Both technology shocks are identified with long run restrictions, which do not pin down the shape of the response on impact, and for this reason some of these responses differ more between the baseline and the aligned version than the monetary policy responses do.

Structural parameters and implied duration

Table 8 compares the DSGE parameters estimated once every VAR is aligned on a common sample with those estimated in the baseline exercise, already reported in Section 5.1. I have not re-run the Monte Carlo exercise on the aligned results: the structural parameters, and especially the implied Calvo price parameters, are very close to the baseline and lie inside its confidence interval, so a separate simulation would add little.

Table 8: DSGE structural parameters: VAR-alignment robustness against the baseline

	VAR alignment			**Benchmark**		
Parameter	$p = 4$	$p = 8$	$p = 12$	$p = 4$	$p = 8$	$p = 12$
	Structural Estimated Frictions					
Price markup (λ_f)	1.0100	1.0100	1.0100	1.0100	1.0100	1.0100
	(-)	(-)	(-)	(-)	(-)	(-)
Wage stickiness (ξ_w)	0.6968***	0.6059***	0.4838***	0.7221***	0.5617***	0.4838***
	(0.1437)	(0.1010)	(0.0757)	(0.1506)	(0.1142)	(0.0757)
Phillips curve slope (γ)	0.0320*	0.0580	0.0129***	0.0402	0.0402	0.0129***
	(0.0184)	(0.0462)	(0.0042)	(0.0246)	(0.0598)	(0.0042)
Capacity util. cost (σ_a)	1.7405	0.1912	0.0042	1.9938	0.1175	0.0042
	(4.5280)	(0.8913)	(0.0230)	(2.6066)	(0.6299)	(0.0230)
Habit formation (b)	0.6956***	0.7117***	0.8596***	0.7066***	0.6935***	0.8596***
	(0.0537)	(0.0454)	(0.0203)	(0.0463)	(0.0570)	(0.0203)
Inv. adj. cost (S'')	3.1915	5.6111	6.8431***	3.2807*	3.0372*	6.8431***
	(2.5697)	(4.6727)	(2.3306)	(1.8874)	(1.5506)	(2.3306)
Money demand elast. (ϵ)	0.8580***	0.9859***	0.8346***	0.8077***	0.7531***	0.8346***
	(0.1943)	(0.2388)	(0.1944)	(0.2111)	(0.1473)	(0.1944)
	Implied Price Stickiness					
Homogeneous capital ($\xi_{p,hom}$)	0.8392***	0.7890***	0.8958***	0.8213***	0.8213***	0.8958***
	[0.0480]	[0.0574]	[0.0139]	(0.0480)	(0.0574)	(0.0139)
Firm-specific capital ($\xi_{p,fs}$)	0.3862**	0.5326***	0.8884***	0.3372**	0.6448***	0.8884***
	[0.1505]	[0.1285]	[0.0220]	(0.1505)	(0.1285)	(0.0220)

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The left block re-estimates the exercise with every VAR aligned on a common sample; the right block reproduces the baseline of Table 2. The Calvo price parameters $\xi_{p,hom}$ and $\xi_{p,fs}$ are not estimated directly: they are recovered numerically from the estimated structural parameters, so their standard errors come from a 1,000,000-draw Monte Carlo of the joint distribution of the estimated structural parameters rather than from the minimum-distance step. That Monte Carlo was run on the baseline estimates only. The figures in square brackets under the VAR-alignment block therefore repeat the baseline Monte Carlo standard errors and are reported purely as a yardstick for the size of the left-right gap; they are not standard errors of the aligned estimates.

The two most notable movements are in γ and σ_a, the Phillips curve slope and the capacity utilisation cost. In the aligned exercise both estimates are smaller in magnitude than in the baseline: $\gamma_{4,\text{aligned}} = 0.0320$ becomes statistically significant at the 10 percent level, where $\gamma_{8,\text{aligned}} = 0.0580$ remains indistinguishable from zero. The capacity utilisation cost becomes $\sigma_{a;4,\text{aligned}} = 1.7405$ and $\sigma_{a;8,\text{aligned}} = 0.1912$, both remaining indistinguishable from zero, as in the baseline at every lag order.

Table 9 translates the implied Calvo parameters into implied durations, aligned against baseline.

Table 9: Implied average time between price re-optimisations: VAR-alignment robustness against the baseline

	VAR alignment			**Benchmark**		
Model specification	$p = 4$	$p = 8$	$p = 12$	$p = 4$	$p = 8$	$p = 12$
Homogeneous capital model						
Calvo parameter ($\xi_{p,hom}$)	0.84	0.79	0.90	0.82	0.82	0.90
Implied duration (quarters)	**6.2**	**4.7**	**9.6**	**5.6**	**5.6**	**9.6**
Firm-specific capital model						
Calvo parameter ($\xi_{p,fs}$)	0.39	0.53	0.89	0.34	0.64	0.89
Implied duration (quarters)	**1.6**	**2.1**	**9.0**	**1.5**	**2.8**	**9.0**

Note: Implied duration is $D = 1/(1 - \xi_p)$, in quarters. The left block re-estimates the exercise with every VAR aligned on a common sample; the right block reproduces the baseline. At $p = 12$ the two coincide by construction, since aligning on the longest lag order leaves that specification unchanged. No standard errors are reported: D is a strongly convex function of ξ_p, so a delta-method interval would be unreliable precisely where ξ_p approaches unity and the Monte Carlo distributions of ξ_p are markedly non-normal.

In the homogeneous capital model the aligned exercise implies a duration almost one quarter longer than the baseline at $p = 4$ ($D_{4,\text{aligned}} = 6.2$ against $D_{4,\text{baseline}} = 5.6$), but almost one quarter shorter at $p = 8$ ($D_{8,\text{aligned}} = 4.7$ against $D_{8,\text{baseline}} = 5.6$). In the firm-specific capital model the implied duration moves by no more than 0.7 quarters, again at $p = 8$, making it the marginally more robust of the two specifications under this particular check.

None of these movements are large enough to change the conclusions drawn from the full-sample baseline in the main text. The collapse of σ_a and the convergence of the implied durations documented there as the lag order rises from $p = 4$ to $p = 12$ is therefore a feature of the lag order itself, not an artefact of the smaller sample that a longer lag order happens to require.